

Robust Narrative Persuasion

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Abstract

This paper studies a sender who designs a public menu of narratives for a heterogeneous population whose cross-sectional distribution of priors is known only through moment restrictions. After each realization, receivers select the narrative that best explains it under their prior and act on its posterior. With binary states, any menu is weakly dominated by a menu using one narrative to set each realization's threshold. I characterize implementable thresholds by a frontier that is piecewise linear in odds coordinates and reduce robust menu design to a moment problem in finitely many dimensions that admits a least favorable distribution with finite support. In a political application, an optimal menu concentrates persuasion on the more valuable realization. Tighter dispersion raises the sender's guarantee, while a higher mean prior can lower it by reducing the least favorable mass above the persuasion threshold.

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1 Introduction

The same public event can support opposing interpretations. Voters may treat an election outcome as a legitimate mandate or as a sign of manipulation; consumers may view a product review as reliable information about quality or as a paid endorsement; jurors may regard forensic evidence as proof of guilt or as procedurally flawed.

Each such interpretation is a *narrative*: a likelihood model that assigns probabilities to possible observations in each state of the world. Each receiver adopts the narrative that best fits what he observes under his prior.¹ Which narrative he chooses depends on what

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¹Throughout, I refer to the sender as she and each receiver as he.

he already believes. After an unfavorable protest report, skeptical citizens may infer regime weakness, while loyal citizens may attribute it to foreign interference.

A sender offering a public menu of narratives must anticipate this self-selection. By designing the menu, she shapes narrative selection. For a given signal realization, persuasion succeeds for a receiver if his selected narrative induces a sender-favorable action. If few receivers lie near the margin of being persuaded, a population with a favorable average prior may still be hard to persuade. Conversely, a population with a lower average prior can be more responsive if its mass is concentrated among the marginal receivers. This makes the cross-sectional distribution of priors central.

In practice, the sender rarely observes the population distribution. I call this uncertainty *ambiguity*. A dictator infers public dissent through rally attendance and social media monitoring; a seller derives consumer attitudes from aggregated market surveys; a prosecutor prepares for jury selection using records of collective past behavior. Such information typically takes the form of sample statistics, such as averages and measures of dispersion. These sample statistics define an *ambiguity set*. The sender evaluates each menu against all distributions in this set; her worst-case expected payoff over this set is the menu’s *payoff guarantee*. The question is: How should she design a public menu of narratives to maximize her payoff guarantee in a heterogeneous population?

To answer this question, I formulate a problem of robust narrative persuasion. The sender searches for an optimal public menu that is *robust* in two senses. First, it performs well against the most unfavorable distribution in the ambiguity set, as if a hypothetical Nature were choosing the distribution adversarially. Second, its payoff guarantee is stable to small perturbations of the ambiguity set: when the ambiguity set is slightly misspecified, the payoff guarantee does not drop discontinuously.

In the paper’s framework, the sender commits ex ante to a public menu of narratives, each explaining the signal realizations produced by the true information structure, which I call the *data-generating model* following Jain (2023).² This departs from Bayesian persuasion (Kamenica and Gentzkow, 2011), where the sender chooses the true information structure.³

The public nature of the menu requires the sender to anticipate varied reactions across receivers; its choice before the realization makes her account for how a receiver’s narrative selection changes across realizations; and her knowledge of only sample statistics about priors requires the menu to perform well for every distribution consistent with those statistics. Therefore, the sender faces a public-menu design problem under ambiguity. The same design problem has an alternative interpretation: for a single receiver, the sender is uncertain both about that receiver’s prior and about the distribution from which it is drawn, knowing only sample statistics of the latter.

The key tradeoff arises around the margin of persuasion. After each signal realization, the menu separates receivers who take the sender-preferred action from those who do not. Given any menu, Nature selects from the ambiguity set the least favorable distribution, placing as much mass as possible among receivers who are not persuaded by that menu. The sender chooses the menu with the highest payoff guarantee. But the models in the menu compete

²By “explain,” I mean specifying a probabilistic relationship between states and signal realizations.

³See Bergemann and Morris (2019) and Kamenica (2019) for surveys of Bayesian persuasion and its subsequent extensions.

on fit after every realization: making narrative more persuasive after one can improve its fit after another and crowd out the narrative intended there.

Under binary states, this logic reduces to a threshold profile. One threshold-setting narrative per realization is enough, and implementability is characterized by a persuasion frontier in threshold-odds coordinates.

I illustrate this tradeoff in an application. There, I assume that whenever a receiver believes it is weakly more likely to be in the state where the sender-preferred action is also the correct choice, he takes the sender-preferred action. Under this assumption, a boundary menu is optimal: after the realization with the larger sender payoff weight, every receiver takes the sender-preferred action; after the other realization, the sender earns payoff only from receivers whose priors are high enough. A tighter lower bound on prior dispersion weakly raises the optimal value, and raises it strictly when this lower bound binds in Nature’s least favorable distribution. A higher average prior can lower the optimal value if Nature can satisfy the higher mean with less mass above the margin of persuasion.

Related literature

This paper builds on the model-as-likelihoods approach to model persuasion introduced by [Schwartzstein and Sunderam \(2021\)](#).⁴ [Aina \(2025\)](#) extends this line to ex ante menu choice mainly for a single receiver with a known prior. I keep the ex ante formulation and extend it to a public menu for a heterogeneous population, studying how partial knowledge of the cross-sectional distribution of priors changes menu design.⁵

The paper’s focus on a common menu connects the analysis to public persuasion. [Laclau and Renou \(2016\)](#) analyze the restrictions prior heterogeneity imposes when a sender influences multiple receivers through a single public signal; [Alonso and Câmara \(2016\)](#) study a voting application; [Inostroza and Pavan \(2025\)](#) study public information design under worst-case adversarial coordination when receivers also possess exogenous private information.⁶ While those papers study standard Bayesian persuasion where the sender commits to a signal structure, the public commitment object here is a menu of narratives, and heterogeneity enters through receivers’ fit-based selection from that menu.

The narrative-menu setting also yields different comparative statics for dispersion from the Bayesian persuasion setting in [Gitmez and Molavi \(2022\)](#): in their model, greater polarization narrows the scope for public information manipulation, whereas here a tighter lower bound on prior variance shrinks Nature’s feasible set and weakly raises the payoff guarantee.

⁴Another approach represents narratives as causal models via directed acyclic graphs ([Spiegler, 2016](#); [Eliaz and Spiegler, 2020](#); [Eliaz et al., 2021](#)).

⁵As noted above, the same design problem has an alternative interpretation for a single receiver: the sender is uncertain both about that receiver’s prior and about the distribution from which it is drawn, knowing only sample statistics of the latter. This interpretation relaxes the known-prior assumption in [Aina \(2025\)](#). I adopt the population interpretation for exposition.

⁶Other multi-receiver public-information work studies different forms of cross-receiver structure: [Bardhi and Guo \(2018\)](#) characterize sender-optimal experiments under unanimous consent with heterogeneous thresholds of doubt; a separate game-theoretic line studies information design with strategic interaction among receivers, e.g., [Mathevet et al. \(2020\)](#). See also [Doval and Smolin \(2024\)](#) on the welfare consequences of information policies in a population setting.

This paper also contributes to robust persuasion by analyzing narrative design under an ambiguous cross-sectional distribution of priors, whereas existing work focuses on information design. [Hu and Weng \(2021\)](#) study uncertainty about the receiver’s private information source; [Dworczak and Pavan \(2022\)](#) study robustness to exogenous information and adversarial strategy selection; [Sapiro-Gheiler \(2024\)](#) studies ambiguity in receiver preferences via a maxmin criterion over cutoff distributions; [Rosenthal \(2025\)](#) develops a data-driven approach in which the sender infers plausible priors from observed action distributions.⁷ [Kosterina \(2022\)](#) is closest, studying an uncertain individual prior under a sender-designed information structure. The paper’s setting differs on two counts. The instrument is different: the data-generating model is fixed, while Kosterina’s sender designs the information structure. The uncertainty is also different: in her paper, the sender hedges against the worst prior in a set, whereas here the sender is uncertain about the cross-sectional distribution of priors.

I apply the robustness criterion and the continuous moment ambiguity of [Ball and Katwink \(2026\)](#); introducing additional structural assumptions provides a tractable way to solve the inner problem. Beyond persuasion, such moment restrictions are widely used in robust mechanism design, including selling, principal-agent, and reserve-pricing problems ([Carrasco et al., 2018, 2019](#); [Suzdaltsev, 2022](#)). This specification further links the paper’s framework to partial identification ([Manski, 2003](#)), where restrictions define an identified set, and sampling uncertainty is captured by estimated moment sets or confidence regions ([Imbens and Manski, 2004](#)).⁸

The paper proceeds as follows. Section 2 introduces narratives, fit-based selection, and moment ambiguity. Section 3 characterizes menu reduction, implementability, and payoff guarantees. Section 4 presents the political-narrative application. Section C presents a product-review application. Proofs are deferred to the [Appendix](#).

2 Model

2.1 Primitives

The state space is $\Omega = \{H, L\}$ and the signal space S is finite. Each receiver chooses an action $a \in \{0, 1\}$, where $a = 1$ denotes the sender-preferred action.

The sender’s payoff from a receiver depends on that receiver’s action and the state, $U_S : \{0, 1\} \times \{H, L\} \rightarrow \mathbb{R}$. A receiver’s payoff depends on his action and the state, $U_R : \{0, 1\} \times \{H, L\} \rightarrow \mathbb{R}$.

Assumption 1 (Sender’s preferred action)

The sender prefers action 1 in every state:

$$U_S(1, H) > U_S(0, H) \quad \text{and} \quad U_S(1, L) > U_S(0, L).$$

This assumption fixes persuasion as inducing action 1. It does not require the sender’s payoff gain from action 1 to be the same across states.

⁷Alternative robustness criteria include regret minimization ([Babichenko et al., 2022](#)) and minimax-loss ([Parakhonyak and Sobolev, 2026](#)).

⁸Set-valued parameters and uncertainty are studied within a random-set framework by [Molchanov and Molinari \(2018\)](#).

The sender has a prior $\tilde{\mu} \in \Delta(\{H, L\})$. Each receiver has a prior $\theta \in [0, 1]$, representing his probability estimation of state H . This paper identifies receiver types with their prior on H and write $\Theta := [0, 1]$ for the type space. Types are drawn from a *cross-sectional distribution* $F \in \Delta(\Theta)$, where $\Delta(\Theta)$ is endowed with the weak topology. In the *benchmark problem*, the sender knows F ; in the *robust problem*, she faces ambiguity about F , as in Section 2.3.

The *data-generating model* $\pi : \Omega \rightarrow \Delta(S)$ is the true model generating signal realizations; it specifies a conditional distribution $\pi(\cdot | \omega) \in \Delta(S)$ for each state ω . This model is perfectly observed by the sender, while receivers only observe the signal realization s .

Assumption 2 (Regularity)

(a) $\tilde{\mu}(H) > 0$ and $\tilde{\mu}(L) > 0$; and (b) for every $s \in S$, $\pi(s | H) > 0$ or $\pi(s | L) > 0$.

Assumption 2 rules out degenerate cases by ensuring both states have positive ex ante weight and every realization can occur, allowing the menu problem to be stated without irrelevant realizations.

2.2 Narratives

A *menu of narratives* is a finite set $M = \{m_1, \dots, m_J\}$.⁹ Each narrative m_j is a map $m_j : \{H, L\} \rightarrow \Delta(S)$, assigning a distribution over realizations to each state. For each integer $J \geq 1$, $\mathcal{M}_J$ denotes the set of all menus with J narratives; the *menu space* is $\mathcal{M} := \bigcup_{J \geq 1} \mathcal{M}_J$, unrestricted relative to the data-generating model π . This convention follows Schwartzstein and Sunderam (2021) and subsequent work.

What governs the realization remains π , while the menu space concerns receivers' feasible interpretations. Receivers do not observe π or construct narratives outside M : the announced menu is the only source of narratives through which they can interpret the realization.¹⁰

For each realization $s \in S$ and each receiver type $\theta \in [0, 1]$, the *fit* of narrative m_j is

$$\text{Fit}(s, m_j, \theta) := (1 - \theta) m_j(s | L) + \theta m_j(s | H), \quad (1)$$

the marginal likelihood of s under narrative m_j given prior θ , so higher fit means that the narrative makes the observed realization more likely for that receiver.

Given $s \in S$, $\theta \in [0, 1]$ and $M \in \mathcal{M}$, each receiver selects

$$m^*(s, \theta) \in \arg \max_{m \in M} \text{Fit}(s, m, \theta).$$

As the fit of each narrative depends on the prior, after a common realization s , receivers with different priors may select different narratives. The menu thus groups types according to the narratives they select.

With a slight abuse of notation, when the realization is s and the type is θ , I write $m^*(s | H)$ and $m^*(s | L)$ for the likelihoods assigned to s in each state by the selected narrative $m^*(s, \theta)$.

⁹I use narrative and model interchangeably only for what the sender offers.

¹⁰Section A.6 presents an extension in which the data-generating model remains available as a default model, following the non-dogmatic model-comparison logic in Ba et al. (2026): a receiver may still select a sender-provided narrative when it better explains the observed realization.

Following narrative selection, whenever $\text{Fit}(s, m^*(s, \theta), \theta) > 0$, the receiver applies Bayes' rule to update his belief. Writing $\hat{\mu}_s(\theta)$ for the posterior probability of state H :

$$\hat{\mu}_s(\theta) := \frac{\theta m^*(s | H)}{(1 - \theta) m^*(s | L) + \theta m^*(s | H)}. \quad (2)$$

Given the posterior, the receiver chooses an action to maximize his expected payoff

$$a^*(s, m^*(s, \theta), \theta) \in \arg \max_{a \in \{0,1\}} [(1 - \hat{\mu}_s(\theta)) U_R(a, L) + \hat{\mu}_s(\theta) U_R(a, H)].$$

Write $a_M(\theta, s) := a^*(s, m^*(s, \theta), \theta)$ for the action of a type- θ receiver after realization s under menu M .

The sender's *per-type payoff*, namely type θ 's contribution to her ex ante payoff, is

$$v(\theta; M) = \tilde{\mu}(H) \sum_{s \in S} \pi(s | H) U_S(a_M(\theta, s), H) + \tilde{\mu}(L) \sum_{s \in S} \pi(s | L) U_S(a_M(\theta, s), L), \quad (3)$$

where U_S is defined in Section 2.1.

Under the cross-sectional distribution F , the sender's payoff from menu M is

$$W(M, F) = \int_{\Theta} v(\theta; M) F(d\theta).$$

The timing is as follows. Types $\theta \in [0, 1]$ are drawn from $F \in \Delta([0, 1])$. The sender, with prior $\tilde{\mu} \in \Delta(\{H, L\})$, commits to a deterministic public menu $M \in \mathcal{M}$, and receivers observe M .¹¹ Nature draws $\omega \in \{H, L\}$ according to $\tilde{\mu}$ and then $s \in S$ according to $\pi(\cdot | \omega)$. Observing s , each receiver selects $m \in M$ maximizing $\text{Fit}(s, m, \theta)$, updates his belief by (2), and chooses an action. Payoffs are realized.

This commitment rules out ex post tailoring: after s is realized, the sender cannot replace or supplement the announced menu and then ask receivers to interpret the realization only through the revised set.

Conventions Tie-breaking is sender-favorable, as in Aina (2025). If multiple narratives maximize fit, the receiver selects one that yields the highest expected payoff for the sender; if both actions maximize his utility, he chooses the sender-preferred one.

2.3 Ambiguity

In the robust problem, the sender only knows that the cross-sectional distribution F lies in $\Pi \subseteq \Delta(\Theta)$, and she evaluates each menu M by $V(M) := \inf_{F \in \Pi} W(M, F)$.

Without further structure on Π , however, a worst-case criterion need not deliver robustness: payoff guarantees may vary discontinuously under small perturbations of Π . To rule

¹¹The sender commits to one deterministic public menu. A public lottery over menus would deliver the mixed problem in Section A.8, which eliminates the max-min gap. I exclude that device because a one-shot public lottery over official frameworks is hard for a dispersed receiver to verify and, even with one receiver, would require him to verify both the announced lottery and the sender's compliance with the realized draw (Fr chet  et al., 2022; Lin and Liu, 2024; Best and Quigley, 2024).

out this instability, the ambiguity set is specified by continuous moment restrictions together with an interiority condition. These restrictions capture what the sender can credibly estimate or calibrate about the distribution.

Definition 1 (Moment ambiguity set)

Fix an integer $d \geq 1$, a continuous vector-valued function $g : \Theta \rightarrow \mathbb{R}^d$, called the moment map, and a non-empty closed convex set $Y \subseteq \text{conv } g(\Theta)$, called the moment set. The moment ambiguity set generated by (g, Y) is

$$\Pi = \mathcal{F}(g, Y) := \{F \in \Delta(\Theta) : \mathbb{E}_{\theta \sim F}[g(\theta)] \in Y\}.$$

I impose the following interiority condition on the moment set $Y \subset \text{conv } g(\Theta)$.¹² The set Y is *star g -interior*: there exists $y_0 \in Y \cap \text{ri}(\text{conv } g(\Theta))$ such that $[y_0, y] \subseteq Y$ for every $y \in Y$. This rules out moment information whose feasible moment set has no relative-interior star center, although Y may still contain boundary points of the feasible moment region. Since the sender's summary statistics may be estimated with small errors, the payoff guarantee should not depend on exact boundary equalities alone. By Ball and Kattwinkel (2026, Propositions 5 and 6(i)), this regularity places Π in their class of continuous moment sets, which are rich and hence uniformly robust. A cross-sectional distribution that is close to Π but slightly misses the moment restrictions can be adjusted back into Π by reallocating only a vanishing amount of probability mass. For distributions near Π , any shortfall below the sender's payoff guarantee is uniformly small across admissible menus.

Example 1 (mean constraints)

Consider a mean constraint, $g(\theta) = \theta$, with $Y = [\underline{\theta}, \bar{\theta}] \subseteq [0, 1]$. The ambiguity set is $\Pi = \{F \in \Delta(\Theta) : \underline{\theta} \leq \mathbb{E}_F[\theta] \leq \bar{\theta}\}$, which collapses to a fixed mean $\Pi = \{F : \mathbb{E}_F[\theta] = \hat{\theta}\}$ when $\underline{\theta} = \bar{\theta} = \hat{\theta}$. Here Y is star g -interior when it intersects $(0, 1)$; among nonempty closed convex subsets of $[0, 1]$, this excludes only the degenerate singletons $Y = \{0\}$ and $Y = \{1\}$.

Example 2 (mean and variance constraints)

Consider a mean constraint and a lower bound on variance: $\Pi = \{F \in \Delta(\Theta) : \mathbb{E}_F[\theta] = \hat{\theta}, \text{Var}_F(\theta) \geq \underline{\sigma}^2\}$, where $0 < \hat{\theta} < 1$ and $0 < \underline{\sigma}^2 < \hat{\theta}(1 - \hat{\theta})$. This set can be represented in moment form by taking $g(\theta) = (\theta, \theta^2)$ and $Y = \{(\hat{\theta}, q) : q \in [\hat{\theta}^2 + \underline{\sigma}^2, \hat{\theta}]\}$, where the lower endpoint imposes the variance lower bound and the upper endpoint follows from $\theta^2 \leq \theta$ on $[0, 1]$. The resulting Y is a vertical segment, and the strict bound $0 < \underline{\sigma}^2 < \hat{\theta}(1 - \hat{\theta})$ ensures that it is star g -interior. Notice that here Y is also closed and convex.

3 Results

Every signal realization generates one persuasion threshold. Since the state space is binary and each receiver has state-matching preference, narrative selection and action choice are cutoff-based: after each $s \in S$, the menu separates types at a threshold θ_s , distinguishing those who take the sender-preferred action from those who do not. Thus a menu matters for

¹²Alternatively, one could allow Y to be larger and impose the condition on the relevant set $Y \cap \text{conv } g(\Theta)$. I adopt the simpler formulation in which $Y \subseteq \text{conv } g(\Theta)$.

payoffs only through the threshold profile $(\theta_s)_{s \in S}$. The main results characterize this profile: one threshold-setting narrative per realization is enough, implementability is governed by cross-realization restrictions on persuasion thresholds, and the robust inner problem is solved for the induced threshold payoff.

3.1 Benchmark

The paper first studies the no-ambiguity problem, in which the cross-sectional distribution of priors F is known. It provides the benchmark for the robust analysis. Even when the population is known, commitment to one menu constrains persuasion across realizations. The sender would like to make each realization persuasive for as many receivers as possible. However, after every realization, receivers choose from the same set of narratives: a narrative that lowers the cutoff after one realization can also become an attractive interpretation after another realization. Hence the payoff-relevant cutoffs cannot be chosen independently.

Receiver types $\theta \in [0, 1]$ are scalar priors on H , as in Section 2.1.¹³

I impose the following assumption on receiver preferences.

Assumption 3 (Receiver preferences)

The receiver has state-matching preferences:

$$u_H := U_R(1, H) - U_R(0, H) > 0 \quad \text{and} \quad u_L := U_R(1, L) - U_R(0, L) < 0.$$

Assumptions 1 and 3 impose a simple conflict of interest: the sender always prefers action 1, regardless of the state, while the receiver prefers action 1 in state H and action 0 in state L .

Two consequences follow. The sender's payoff depends on the menu only through the induced action rule, and the receiver chooses $a = 1$ if and only if his posterior on H is at least the *action threshold* $\tau := |u_L|/(u_H + |u_L|) \in (0, 1)$. Both action choice and narrative selection are then cutoffs in type space, collapsing the problem of choosing the optimal menu to the choice of persuasion thresholds defined next.

Persuasion thresholds and sender payoff structure. Under binary states, each signal realization $s \in S$ induces one endogenous persuasion threshold. For a fixed realization s , fit is affine in θ , with slope $m(s | H) - m(s | L)$. Along the upper envelope of these fit functions, selected slopes are nondecreasing in type, and sender-favorable tie-breaking selects the higher posterior at fit switches. Hence the set of types persuaded after s is an upper interval.

Definition 2 (Persuasion threshold)

Fix a menu $M \in \mathcal{M}$. For each realization $s \in S$ and type $\theta \in [0, 1]$, let $m^(s, \theta)$ be the narrative selected from M by the fit-based rule, and write $\hat{\mu}_s(\theta) := \mu_s^{m^*(s, \theta)}(\theta) \in [0, 1]$ for*

¹³Prior heterogeneity can also summarize other forms of heterogeneity in receiver behavior. With a common prior and private signals observed before the public realization, receivers' interim beliefs can serve as types for narrative selection and action. Under Assumption 3, a setting with a homogeneous prior and heterogeneous action thresholds can likewise be represented, for any fixed selected narrative, by heterogeneous effective priors.

the posterior of state H obtained by updating under the selected narrative. The persuasion threshold after realization s under menu M is the type-space cutoff

$$\theta_s(M) := \inf\{\theta \in [0, 1] : \hat{\mu}_s(\theta) \geq \tau\}, \quad s \in S, \quad (4)$$

with the convention $\theta_s = 1^+$ if the corresponding set is empty, where 1^+ is an artificial point above 1 and $\mathbf{1}\{\theta \geq 1^+\} = 0$ for every $\theta \in [0, 1]$.

When the set defining θ_s is nonempty, the threshold θ_s is the marginal type after s . Types above θ_s are those whose selected narrative induces a posterior of at least τ , while types below θ_s are not persuaded. When $\theta_s = 1^+$, no type is persuaded after s . A lower θ_s means that the menu persuades more types after s .

Because the payoff $v(\theta; M)$ depends on M only through the threshold profile $(\theta_s(M))_{s \in S}$, define

$$\bar{u}_0 := \sum_{\omega} \tilde{\mu}(\omega) U_S(0, \omega), \quad w_s := \sum_{\omega} \tilde{\mu}(\omega) \pi(s | \omega) [U_S(1, \omega) - U_S(0, \omega)], \quad s \in S.$$

By Assumptions 1 to 3, $w_s > 0$ for every $s \in S$. I call w_s the sender's *payoff weight* for realization s : it is her ex ante payoff increment from inducing one additional receiver to take action 1 after s . The constant $\bar{u}_0$ is her ex ante payoff when no receiver is persuaded; unlike the payoff weights, it does not depend on the thresholds. The profile $(\theta_s)_{s \in S}$ induces the per-type payoff

$$v(\theta; (\theta_s)_{s \in S}) = \bar{u}_0 + \sum_{s \in S} w_s \mathbf{1}\{\theta \geq \theta_s\}. \quad (5)$$

The artificial threshold convention makes the corresponding indicator in (5) the zero function when the persuasion set is empty. I then define

$$V((\theta_s)_{s \in S}) := \inf_{F \in \Pi} \int_{\Theta} v(\theta; (\theta_s)_{s \in S}) F(d\theta), \quad (6)$$

and set $V(M) := V((\theta_s(M))_{s \in S})$. The benchmark problem is the singleton case $\Pi = \{F\}$, where (6) specializes to $\int v dF$; the robust problem in Section 3.2 uses a non-singleton Π .

For each realization with a nontrivial threshold, call a narrative *threshold-setting* if the marginal type selects it and reaches a posterior of at least τ . The selection convention in Section 2.2 is structural for the menu-reduction argument: after each realization, it implies that the types who are persuaded form an upper interval with cutoff θ_s .

Threshold-setting narratives simplify the problem by reducing the menu choice set from the whole menu space $\mathcal{M}$ to a smaller class. Keeping only one threshold-setting narrative for each signal realization, each retained narrative remains fit-maximal for its original marginal type, and, since the posterior induced by that narrative is increasing in θ , it also persuades all types above the original threshold after that realization.

Theorem 1 (Menu reduction)

Under binary states and any finite signal space S , for every menu M , there exists a sub-menu $M' \subseteq M$ with $|M'| \leq |S|$ such that $v(\theta; M') \geq v(\theta; M)$ for every $\theta \in [0, 1]$, and hence $V(M') \geq V(M)$.

The reduction shows that there is no loss in restricting attention to the narratives that set the realization-specific persuasion thresholds. An additional narrative that does not set such a threshold can matter only by changing which narrative some receivers select. It can reduce persuasion when it wins the fit comparison for some receivers while giving them a posterior below the action threshold.

Example 3 (Removing a non-threshold-setting narrative)

Let $M = \{(1/8, 3/8), (1/3, 1/3), (1, 0)\}$ and $\tau = 1/2$. After h , types on $[1/6, 1/3)$ select the intermediate narrative from the full menu and switch to $(1, 0)$ only at $1/3$, so $\theta_h(M) = 1/3$; after l , the posterior under $(1/8, 3/8)$ crosses τ at $5/12$, giving $\theta_l(M) = 5/12$. The sub-menu $M' = \{(1/8, 3/8), (1, 0)\}$ has a fit-crossing point $3/10$: types below $3/10$ select $(1/8, 3/8)$ and have posterior $\theta/(3-2\theta) < 1/2$ after h , while types above $3/10$ select $(1, 0)$ and have posterior 1. Hence $\theta_h(M') = 3/10 < 1/3 = \theta_h(M)$, with θ_l unaffected since the l -side selection does not involve the removed intermediate narrative. The intermediate narrative wins on fit but yields lower posteriors; its presence delays persuasion to higher types. This illustrates Theorem 1: removing the narrative that was delaying persuasion after h lowers θ_h without affecting θ_l .

Implementability. A threshold profile $(\theta_s)_{s \in S} \in [0, \tau]^{|S|}$ is *implementable* if some menu induces θ_s as the persuasion threshold after each realization $s \in S$.¹⁴ The next result characterizes which threshold profiles can arise from a common menu.

The cross-realization selection constraint arises from fit competition among narratives. The sender uses one public menu after every realization, so a narrative that makes action 1 easier to induce after one realization may also become an attractive interpretation after another realization. Hence the realization-specific persuasion thresholds cannot be chosen independently. Implementability asks which threshold profiles can be generated while each threshold-setting narrative remains selected at its own marginal type.

Theorem 2 (Implementability)

Under binary states and signal space S , the following statements hold.

- (i) A threshold profile $(\theta_s)_{s \in S} \in [0, \tau]^{|S|}$ is implementable by some menu if and only if

$$\sum_{s \in S} \frac{\theta_s}{1 - \theta_s} \geq \tau \left(1 + \max_{s \in S} \frac{\theta_s}{1 - \theta_s} \right). \quad (7)$$

The endpoint $\theta_s = 0$ is read in the right limit.

- (ii) In the odds-ratio coordinates $r_s := \theta_s/(1 - \theta_s) \in [0, \tau/(1 - \tau)]$, the implementable set (7) is the intersection of the $|S|$ halfspaces $\{\sum_{s' \in S} r_{s'} \geq \tau(1 + r_s)\}$, one per $s \in S$. For any fixed ordering $r_{(1)} \leq \dots \leq r_{(|S|)}$, the corresponding piece of the persuasion frontier is

$$\sum_{i=1}^{|S|-1} r_{(i)} + (1 - \tau) r_{(|S|)} = \tau. \quad (8)$$

¹⁴Higher thresholds are dominated: replacing any threshold above τ with τ weakly expands persuasion and cannot strictly reduce the sender's payoff. That is why I focus on $[0, \tau]^{|S|}$. The formal statement is Lemma A.5.

The frontier is the union of the corresponding hyperplanes over all permutations of the signal labels.

In odds coordinates, the inequality has a direct interpretation. A lower persuasion threshold θ_s means stronger persuasion after realization s , and hence lower threshold odds $r_s = \theta_s/(1 - \theta_s)$. The left-hand side aggregates the total threshold burden across realizations. The right-hand side is governed by the largest threshold odds, which corresponds to the tightest cross-realization selection constraint. Thus the menu can lower persuasion thresholds below the action threshold τ , but it cannot lower all of them too much at once.

Corollary 1 (Symmetric persuasion across realizations)

For $t \in [0, \tau]$, the symmetric profile $(t, \dots, t)$ is implementable if and only if $t \geq \tau/|S|$. Consequently, narrative persuasion can symmetrically lower every persuasion threshold below the action threshold τ , but not below $\tau/|S|$.

Corollary 2 (Optimal ordering on the persuasion frontier)

Relabel the signal realizations as $s_1, \dots, s_n$, $n = |S|$, so that $w_{s_1} \geq \dots \geq w_{s_n}$. Every implementable threshold profile is pointwise weakly dominated in v by an implementable profile $(\theta'_{s_i})_{i=1}^n$ with

$$0 \leq \theta'_{s_1} \leq \dots \leq \theta'_{s_n} \leq \tau$$

whose odds coordinates $r'_{s_i} := \theta'_{s_i}/(1 - \theta'_{s_i})$ satisfy the frontier equation (8) under this ordering. Hence the sender's outer problem can be restricted to frontier profiles that assign weakly lower thresholds to weakly higher payoff weights.

The dominance in Corollary 2 has two steps. First, thresholds can be ordered so that lower thresholds are assigned to higher payoff weights. Since a lower threshold persuades weakly more types after that realization, this rearrangement weakly raises the induced payoff function. Second, scaling all threshold odds of the ordered profile down to the boundary in Theorem 2 lowers every threshold, which again weakly raises the induced payoff function. The ordering step rests on a pairwise swap argument; the contraction to the frontier applies Theorem 2 together with the pointwise monotonicity of the payoff in thresholds.

The frontier represents a minimum threshold-burden condition. Under this payoff-weight ordering, lowering the threshold odds of a high-payoff realization requires assigning the remaining burden to lower-payoff realizations. This is the sense in which the sender's search can be restricted to frontier profiles with lower thresholds assigned to higher-payoff realizations.

Proposition 1 (Existence of the optimal menu)

There exists an optimal menu $M^ \in \mathcal{M}$ attaining $V(M^*) = \max_{M \in \mathcal{M}} V(M)$.*

Given Proposition 1, I replace sup with max for the sender's benchmark problem and robust outer problem.

For the figures and examples, write $S = \{h, l\}$ and let (θ_h, θ_l) denote the threshold profile. Specializing Theorem 2 gives the two-dimensional implementability region. For any $\tau \in (0, 1)$, write $\alpha(\theta) := \theta(1 - \tau)/((1 - \theta)\tau)$. A threshold pair $(\theta_h, \theta_l) \in [0, \tau]^2$ is implementable if and only if

$$\alpha(\theta_h) + \alpha(\theta_l) \geq \frac{1 - 2 \min\{\theta_h, \theta_l\}}{1 - \min\{\theta_h, \theta_l\}}. \quad (9)$$

Indeed, if $\theta_h \leq \theta_l$, then (7) becomes $r_h + (1 - \tau)r_l \geq \tau$, which is equivalent to (9); the case $\theta_l \leq \theta_h$ is symmetric. Figure 1 illustrates the implementability region and its frontier.

Define the *implementability slacks*

$$\psi(\theta_h, \theta_l) := \alpha(\theta_h) + \alpha(\theta_l) - \frac{1 - 2\theta_h}{1 - \theta_h}, \quad \phi(\theta_h, \theta_l) := \alpha(\theta_h) + \alpha(\theta_l) - \frac{1 - 2\theta_l}{1 - \theta_l}. \quad (10)$$

On $\theta_h \leq \theta_l$, (9) is equivalent to $\psi \geq 0$; on $\theta_l \leq \theta_h$, it is equivalent to $\phi \geq 0$. Thus

$$\{0 \leq \theta_h \leq \theta_l \leq \tau : \psi \geq 0\}$$

is the *h-first implementability region*, and

$$\{0 \leq \theta_l \leq \theta_h \leq \tau : \phi \geq 0\}$$

is the *l-first implementability region*. The binding portions $\psi = 0$ and $\phi = 0$ are the *h-first branch* and the *l-first branch* of the persuasion frontier. On these branches, write

$$\theta_l = \theta_l(\theta_h), \quad \theta_h \in [0, \tau/2], \quad \theta_h = \theta_h(\theta_l), \quad \theta_l \in [0, \tau/2]. \quad (11)$$

Write $r_s := \theta_s/(1 - \theta_s)$. Substituting $\alpha(\theta_s) = r_s(1 - \tau)/\tau$ into $\psi = 0$ gives the *h-first branch*; the *l-first branch* is symmetric:

$$r_h + (1 - \tau)r_l = \tau \quad (h\text{-first branch}), \quad (1 - \tau)r_h + r_l = \tau \quad (l\text{-first branch}). \quad (12)$$

Figure 1 plots the implementability regions in panel (a) and the linear threshold-burden frontiers in panel (b).

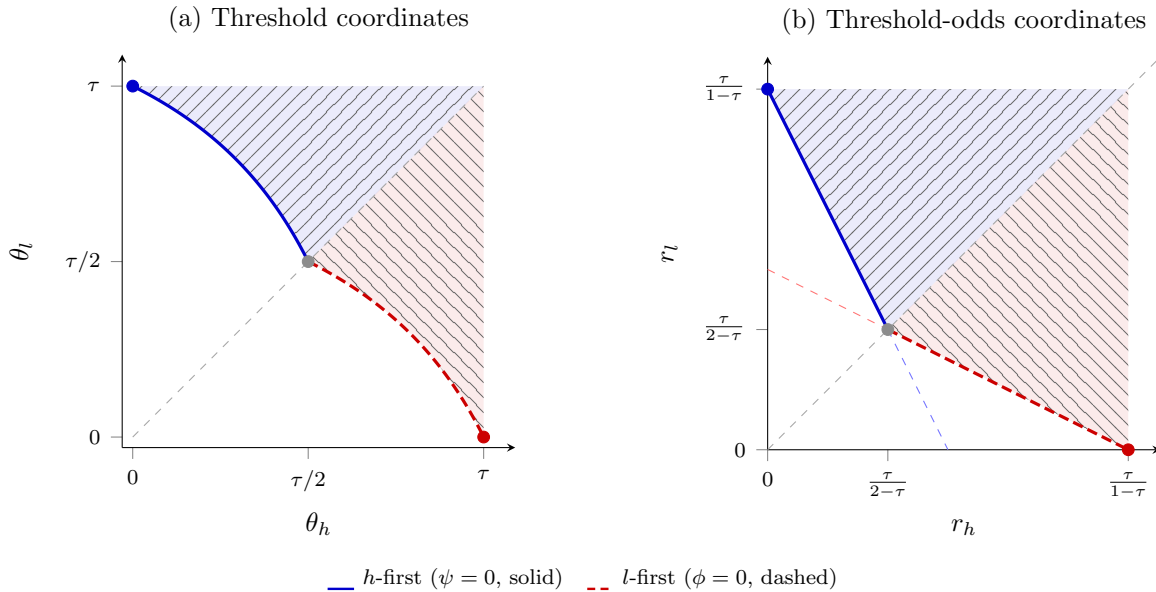


Figure 1: Implementability region and persuasion frontier restricted to $(\theta_h, \theta_l) \in [0, \tau]^2$ ($\tau = 1/2$). Panel (a) uses threshold coordinates (θ_h, θ_l) ; panel (b) uses threshold-odds coordinates (r_h, r_l) , where $r_s = \theta_s/(1 - \theta_s)$. Hatching distinguishes the *h-first* and *l-first* regions; solid and dashed curves mark their frontiers. Dots mark the junction and boundary menus.

3.2 Robustness

With a known population distribution, the sender chooses where to place the threshold burden on the frontier: the weights $(w_s)_{s \in S}$ determine the payoff from persuading receivers after each realization, while the distribution determines the tail mass above each induced threshold. With ambiguity, this tail mass is no longer fixed. For any menu, the sender's payoff also depends on how much receiver mass Nature can keep at or below the induced positive thresholds, subject to the ambiguity set. Given any feasible threshold profile, Nature chooses a distribution that satisfies the moment restrictions and minimizes the sender's payoff. This is the inner problem. The sender evaluates each menu by its payoff guarantee, and then chooses a menu that maximizes this guarantee. This is the outer problem. As in Section 3.1, the outer choice can be restricted to frontier profiles that assign weakly lower thresholds to weakly higher payoff weights. Hence the analysis turns to the inner problem.

Payoff-guarantee equivalence. A receiver type exactly at a positive persuasion threshold cannot support a robust guarantee: under moment ambiguity, Nature can move that mass arbitrarily close to the threshold from below, where the receiver is no longer persuaded. Therefore, at a positive threshold, the sender obtains the payoff increment only from the actions of types strictly above the threshold. At a zero threshold, she also obtains the increment from the action of type $\theta = 0$, because $\Theta = [0, 1]$ leaves Nature no strictly lower type from which to approach zero. This observation motivates evaluating the inner problem with the lower semi-continuous envelope of the per-type payoff:

$$\tilde{v}(\theta; (\theta_s)_{s \in S}) = \bar{u}_0 + \sum_{s \in S} w_s \left(\mathbf{1}\{\theta > \theta_s\} + \mathbf{1}\{\theta_s = 0\} \mathbf{1}\{\theta = 0\} \right), \quad w_s > 0. \quad (13)$$

If $\theta_s = 1^+$, both indicators for realization s are interpreted as zero. Receiver behavior remains as specified in Section 2.2; the envelope is used only to evaluate the inner problem and to obtain an attained worst-case distribution. For a menu M inducing thresholds $(\theta_s(M))_{s \in S}$, define

$$\widetilde{W}(M, F) := \int_{\Theta} \tilde{v}(\theta; (\theta_s(M))_{s \in S}) F(d\theta), \quad \widetilde{V}(M) := \inf_{F \in \Pi} \widetilde{W}(M, F).$$

The function $\tilde{v}$ is lower semi-continuous on Θ , so each menu has an attained payoff guarantee over Π . The next lemma shows that, under the richness of Π , this pointwise convention does not affect the payoff guarantee, the set of robust optimal menus, or the optimal value.

Lemma 1 (Payoff-guarantee equivalence)

Suppose $\Pi = \mathcal{F}(g, Y)$ with Y star g -interior. For each menu $M \in \mathcal{M}$,

$$V(M) = \inf_{F \in \Pi} W(M, F) = \inf_{F \in \Pi} \widetilde{W}(M, F) = \min_{F \in \Pi} \widetilde{W}(M, F).$$

In addition, $\arg \max_M V(M) = \arg \max_M \widetilde{V}(M)$.

Hence, for any menu M , the sender may replace $W(M, F)$ with $\widetilde{W}(M, F)$ before minimizing over Π ; this leaves the payoff guarantee $\inf_{F \in \Pi} W(M, F)$ unchanged, ensures that the minimum is attained, and delivers the finite-support structure used in Proposition 2.

Corollary 3 (Robust frontier restriction)

Order the signal realizations as $s_1, \dots, s_n$, $n = |S|$, so that $w_{s_1} \geq \dots \geq w_{s_n}$. Then

$$\max_{M \in \mathcal{M}} V(M) = \max_{\substack{0 \leq r_{s_1} \leq \dots \leq r_{s_n} \leq \tau/(1-\tau) \\ \sum_{i=1}^{n-1} r_{s_i} + (1-\tau)r_{s_n} = \tau}} \min_{F \in \Pi} \int_0^1 \tilde{v}(\theta; (\theta(r_s))_{s \in S}) F(d\theta),$$

where $\theta(r) := r/(1+r)$.

Combining Proposition 1, Lemma 1, and Corollary 3, the robust problem becomes

$$\max_{M \in \mathcal{M}} \min_{F \in \Pi} \widetilde{W}(M, F),$$

with Nature's minimum attained and the sender's outer choice restricted to the persuasion frontier. For each threshold profile, the sender's payoff guarantee equals what she would earn against the least favorable $F \in \Pi$. As Π expands, the same menu must perform against a broader range of populations, so its payoff guarantee weakly decreases. It remains to characterize Nature's inner minimization problem for a fixed implementable threshold profile.

The inner problem. For each implementable threshold profile, the inner problem identifies the worst-case distribution in the ambiguity set and the associated payoff guarantee, as if Nature were placing receiver mass to minimize the sender's payoff subject to the moment restrictions. Under the maintained assumptions on Π , this problem admits a finite-dimensional dual characterization, paralleling the approach in Carrasco et al. (2018). Under $\tilde{v}$, at a positive persuasion threshold, only mass strictly above the threshold contributes to the sender's persuasion payoff; at a zero persuasion threshold, mass at type $\theta = 0$ also contributes. Thus, for a fixed threshold profile, the inner problem minimizes a weighted sum of upper-tail masses subject to the moment restrictions. The payoff guarantee depends on how those moment restrictions constrain Nature's ability to keep mass at or below the relevant persuasion thresholds. The finite-dimensional dual characterized below gives its value, and a worst-case distribution can be taken to have finite support contained in the contact set between $\tilde{v}(\cdot; (\theta_s)_{s \in S})$ and the dual minorant. Proposition A.4 records the corresponding zero-positive criterion: for a fixed threshold profile, the guarantee above $\bar{u}_0$ is strictly positive when the moment restrictions rule out distributions supported weakly below the lowest positive threshold. This criterion is qualitative; the dual result below gives the level of the guarantee and the structure of least favorable distributions.

Proposition 2 (Worst-case distributions for an implementable threshold profile)

Suppose $\Pi = \mathcal{F}(g, Y)$, where $g : [0, 1] \rightarrow \mathbb{R}^d$ is continuous and $Y \subseteq \text{conv } g([0, 1])$ is nonempty, closed, convex, and star g -interior. Fix an implementable threshold profile $(\theta_s)_{s \in S}$ and evaluate it by the lower semi-continuous payoff $\tilde{v}(\theta; (\theta_s)_{s \in S})$. Then

$$\begin{aligned} \min_{F \in \Pi} \int_0^1 \tilde{v}(\theta; (\theta_s)_{s \in S}) F(d\theta) = & \sup_{(\gamma, \lambda) \in \mathbb{R} \times \mathbb{R}^d} \left\{ \gamma + \inf_{y \in Y} \lambda^\top y : \right. \\ & \left. \gamma + \lambda^\top g(\theta) \leq \tilde{v}(\theta; (\theta_s)_{s \in S}) \text{ for every } \theta \in [0, 1] \right\}. \end{aligned}$$

The minimum is attained, and at least one minimizer can be chosen with at most $d + 1$ support points. For every dual maximizer (γ^*, λ^*) , every worst-case distribution F^* satisfies $\mathbb{E}_{F^*}[g(\theta)] \in \arg \min_{y \in Y} (\lambda^*)^\top y$ and

$$\text{supp}(F^*) \subseteq \{\theta \in [0, 1] : \gamma^* + (\lambda^*)^\top g(\theta) = \tilde{v}(\theta; (\theta_s)_{s \in S})\}.$$

If $g(\theta) = (\theta, \theta^2, \dots, \theta^d)$, then $\gamma^* + (\lambda^*)^\top g(\theta)$ is a polynomial of degree at most d .

In Proposition 2, convexity and star g -interiority of Y give duality, and closedness gives attainment; the robustness result of Ball and Kattwinkel (2026) only requires star g -interiority. The dual problem chooses, from the affine span of the moment functions, a lower-bound function for the lower semi-continuous envelope of the sender's per-type payoff. The infinite-dimensional choice of a distribution is thereby reduced to a finite-dimensional choice of multipliers, i.e. the coefficients (γ, λ) of this affine minorant, with worst-case mass supported only on contact types.

The contact set identifies the receiver types that are most relevant for a proposed menu. These locations are where Nature can place mass while meeting the moment restrictions and keeping the sender's payoff as low as possible. Depending on the moment restrictions and the chosen thresholds, this mass may lie far below the persuasion margin, at the margin, or above it when the restrictions require enough high-prior mass.

Proposition 2 has an equivalent convexification interpretation. For d -moment restrictions, consider the closed convex hull of

$$\{(g(\theta), \tilde{v}(\theta; (\theta_s)_{s \in S})) : \theta \in [0, 1]\} \subseteq \mathbb{R}^{d+1}.$$

Nature's value is obtained from the lower boundary of this set over moment vectors in Y . When $d = 1$, this lower boundary can be read from the figure. Figure 2 illustrates the case with $g(\theta) = \theta$ and $Y = [\underline{\theta}, \bar{\theta}]$. The dual constraint requires $\gamma + \lambda^\top g(\theta)$ to lie below the

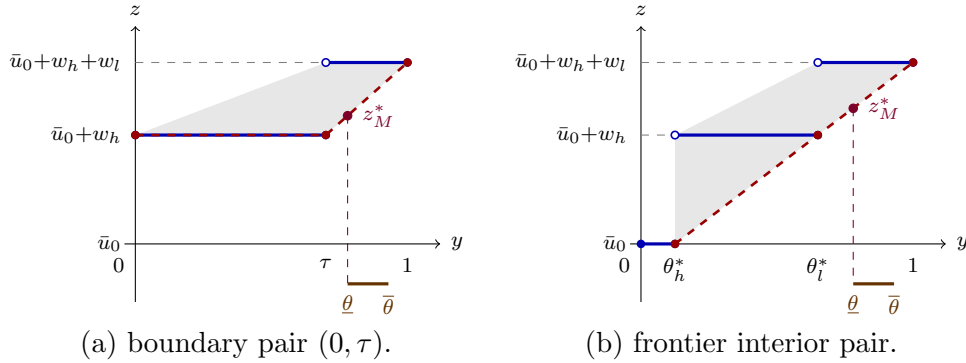


Figure 2: Convexification geometry for $g(\theta) = \theta$, $Y = [\underline{\theta}, \bar{\theta}]$, and the h -first frontier. The shaded region is the closed convex hull of the graph of $\tilde{v}(\cdot; \theta_h, \theta_l)$; the dashed red curve is its lower boundary over Y . Panel (a) shows the boundary threshold pair $(0, \tau)$, and panel (b) shows a frontier interior threshold pair.

lower semi-continuous per-type payoff $\tilde{v}(\cdot; (\theta_s)_{s \in S})$.¹⁵ At each positive threshold, the payoff

¹⁵Because $\tilde{v} \leq v$ pointwise, any affine minorant of $\tilde{v}$ also lies below the original per-type payoff v ; the two functions differ only at positive persuasion thresholds.

takes its left-limit value, while a zero threshold includes type 0 in the persuaded set. Figure 3 illustrates the resulting polynomial minorant under power moments for $d = 3$. The

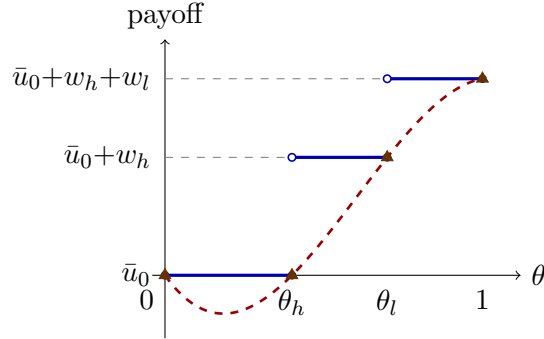


Figure 3: Dual lower bound under power moments. The dashed red polynomial $\gamma + \lambda^\top g(\theta)$ lies below the lower semi-continuous per-type payoff $\tilde{v}(\theta; \theta_h, \theta_l)$ (solid blue), and the worst-case mass concentrates at contact points (orange triangles). One worst-case distribution can be chosen with at most $d + 1$ atoms.

contact-set logic is related to Carrasco et al. (2018, Lemma 2 and Proposition 3): with the first N moments of buyer valuations, the optimal transfer is the non-negative monotonic hull of a degree- N polynomial with coefficients $(\lambda_0, \dots, \lambda_N)$, and Nature is supported on the contact set where transfer and polynomial coincide. Suzdaltsev (2022), by contrast, faces a nonconvex ambiguity set due to the independence constraint on joint value distributions and uses an indirect argument. Finally, Ball and Kattwinkel (2026, Proposition 4) show that a finite-support saddle prior with strictly positive payoff is non-robust in a profit-maximization problem: because the mechanism is tailored to the saddle prior’s atoms, small perturbations can lower payoff discontinuously. Here finite support has a different role: it is an extremal representation of the moment problem, and robustness comes from the richness of the continuous moment ambiguity set.

Does ambiguity change menu choice? The sender’s problem under ambiguity assigns each menu the payoff it generates under its least favorable distribution. The question is whether ambiguity also changes the menu the sender chooses. If a robust-optimal menu has a finite-support worst-case distribution, its payoff may be flat as thresholds move between adjacent support points; then the same menu may already maximize payoff under that distribution. The example below shows the other possibility: a robust-optimal menu lies outside the benchmark solution set at its own least favorable distribution, so ambiguity changes which menu is chosen as well.

Example 4 (Failure of benchmark optimality)

Take $\tau = 1/2$, $\bar{u}_0 = 0$, $w_h = 2$, $w_l = 1$, and $\Pi = \{F \in \Delta([0, 1]) : \mathbb{E}_F[\theta] = 0.4\}$. On the h -first frontier, $\theta_l(x) = (1 - 3x)/(2 - 4x)$ for $x \in [0, 1/4]$. The boundary menu $M_{h\text{-bd}}$ has payoff guarantee 2. For any $x \in (0, 1/4]$, Nature can put mass $(1 - 0.4)/(1 - x)$ at x and mass $(0.4 - x)/(1 - x)$ at 1, giving the frontier menu $(x, \theta_l(x))$ payoff at most $3(0.4 - x)/(1 - x) < 2$. Hence $M_{h\text{-bd}}$ is robust optimal. At $F^* = \delta_{0.4}$, however, any h -first frontier point $(x, \theta_l(x))$ with $x \in (1/7, 1/4]$ satisfies $\theta_l(x) < 0.4$, so type 0.4 is persuaded after both realizations and

yields benchmark payoff $w_h + w_l = 3$, while the boundary menu $(0, 1/2)$ yields only $w_h = 2$. Hence $M_{h\text{-bd}}$ is not benchmark-optimal at F^* .

At $\delta_{0.4}$, the benchmark solution set is the continuum of h -first frontier menus that persuade type 0.4 after both realizations. The boundary menu $M_{h\text{-bd}}$ is robust optimal because it delivers a larger payoff guarantee than those benchmark optima. Hence the robust and benchmark solution sets are not nested. Robustness selects a benchmark optimum only under a saddle-point condition: when the robust-optimal menu and its worst-case distribution are mutual best responses, the menu belongs to both solution sets. The next proposition records this, and Proposition A.5 in Appendix A.12 gives the corresponding criterion.

Proposition 3 (Saddle-point selection from benchmark optima)

If $(M^, F^*) \in \mathcal{M} \times \Pi$ is a saddle point of $\widetilde{W}$, then $M^* \in \arg \max_{M \in \mathcal{M}} \widetilde{W}(M, F^*)$, $M^* \in \arg \max_{M \in \mathcal{M}} \min_{F \in \Pi} \widetilde{W}(M, F)$, and $M^* \in \arg \max_{M \in \mathcal{M}} \inf_{F \in \Pi} W(M, F)$. Robustness selects an element of the $\widetilde{W}$ -benchmark solution set at F^* .*

Proposition 3 implies that determining whether a robust solution is also benchmark-optimal at its worst-case distribution requires checking for profitable deviations. This requires that no shift along the relevant persuasion frontier improves the sender's payoff.

3.3 Discussion

Menu-size bound. When priors are scalar and action is cutoff-based, each realization has a single payoff-relevant marginal type. Retaining one threshold-setting narrative for each realization is enough. A menu with more narratives than signal realizations can always be replaced by an optimal sub-menu with at most as many narratives as signal realizations. Theorem 1 gives this $|S|$ -narrative bound. Richer narratives still affect how receivers explain a signal realization and which types select which narrative, but not in a strictly payoff-improving way.

This differs from the single-receiver size argument in Aina (2025), despite the same $|S|$ bound. There, a single receiver's prior is fixed, so each signal realization can be assigned a narrative that, for that prior, induces the target posterior and attains the maximal feasible fit conditional on that realization. Here, maximal fit is type-dependent: a narrative that maximizes fit for one prior need not maximize fit for another. The sender must control fit-based selection over the whole prior support. The same $|S|$ bound survives because the binary-state environment makes priors scalar and action cutoff-based: controlling selection over priors reduces to retaining one threshold-setting narrative per realization.

One might suspect that the finite-support representation of the worst-case distribution yields another menu-size bound: if a worst-case distribution can be chosen with at most $d + 1$ support points, perhaps only $d + 1$ receiver types need be controlled, and hence $d + 1$ narratives suffice. This inference is not valid. The finite-support result is conditional on a fixed menu, or equivalently on a fixed threshold profile. Its support points are contact types of that menu's inner moment problem, not a menu-independent set of marginal receivers. As the menu changes, the least favorable distribution and its contact set can change as well. Without a common finite contact set that works for all feasible menus and deviations, a support-size bound for the worst-case distribution does not translate into a menu-size bound for narratives.

Beyond binary states. The binary-state assumption makes the receiver’s posterior one-dimensional and, together with a monotone comparison structure in receiver behavior, yields a threshold-choice representation. With more than two states, the same receiver-side representation is recovered when narrative selection and action choice depend on posterior beliefs only through a scalar statistic.

One case is a binary partition. Let H denote the event where the sender-preferred action is also receiver-preferred, let L denote its complement, and denote the action by 1. For instance, let $\nu \in \Delta(Q)$ denote the receiver’s posterior over product quality. If the receiver’s payoff from purchase depends only on whether product quality exceeds an adequacy cutoff q^* , the partition $H = \{q \geq q^*\}$ and $L = \{q < q^*\}$ is sufficient for receiver behavior whenever narrative selection and action choice depend on ν only through $\nu(H)$.

Another case uses a scalar posterior statistic. [Kamenica and Gentzkow \(2011, Proposition 3\)](#) study whether persuasion benefits the sender when her induced payoff depends only on the expected state. Their Proposition 6 extends this criterion to a finite-dimensional linear statistic of posterior beliefs. Under the linear-in-state utilities of [Kolotilin et al. \(2017\)](#), posterior means determine both receiver behavior and the sender’s objective, and messages can be indexed by cutoff types. [Vohra et al. \(2023\)](#) obtain a sender-side reduced form when receiver preferences depend on the posterior mean and sender payoffs depend only on the receiver’s action. [Kolotilin and Zapechelnyuk \(2025\)](#) likewise study linear persuasion problems in which both players’ expected utilities depend on posterior beliefs through the posterior mean.

More generally, a scalar statistic represents receiver behavior when narrative selection and action choice depend on posterior beliefs only through that statistic. It also represents sender behavior when her menu ranking also ignores residual posterior distinctions, and it represents her cardinal payoff under the condition stated in [Appendix A.7](#).

Narrative persuasion versus Bayesian persuasion. In the setting studied here, both narrative persuasion and Bayesian persuasion organize persuaded types into intervals. Using the binary-signal case as an example, on the h -first branch, a boundary menu induces two intervals: types persuaded after both realizations and types persuaded only after h ; a frontier-interior menu induces three intervals: types persuaded after both realizations, types persuaded only after h , and types persuaded after neither realization. The difference lies in the source of the cross-realization restriction. In Bayesian persuasion, Bayes plausibility makes belief movements balance across realizations: a receiver’s posterior cannot lie strictly above his prior after every realization. Narrative persuasion does not impose this balance beyond fit competition. Because receivers may select different narratives after different realizations, narrative persuasion can induce receivers with some priors to choose the sender-favorable action after all realizations.

However, this does not imply that narrative persuasion gives the sender a higher value than Bayesian persuasion, since the data-generating model used to compute the sender’s expected payoff is fixed in narrative persuasion but chosen in Bayesian persuasion. Receiver welfare can move in either direction: he gains when he chooses $a = 1$ more often in state H , and loses when he does this more often in L . For a receiver with prior θ , additional persuasion changes welfare by $\theta u_H \sum_s \pi(s | H) \Delta a_s(\theta) - (1 - \theta) |u_L| \sum_s \pi(s | L) \Delta a_s(\theta)$. A

given receiver gains when the prior-weighted gain in state H exceeds the prior-weighted loss in state L ; if the inequality is reversed, he loses. Because the sign of this welfare change varies with θ , additional persuasion can also raise welfare for some receivers and lower it for others.

4 Application

Consider a dictator who wants citizens to support her ($a = 1$), while citizens only want to support her when she is competent ($\omega = H$). Citizens differ in their prior θ that the dictator is competent. Public news generates a favorable realization h and an unfavorable realization l , with $0 < \pi_L < \pi_H < 1$, so each realization is informative but imperfect. The action threshold is $\tau = 1/2$, as in much of the persuasion literature: a receiver takes the sender-preferred action whenever the state aligned with that action is at least as likely as the alternative (Kamenica and Gentzkow, 2011; Bergemann and Morris, 2019; Kamenica, 2019).

I normalize the dictator's utility when citizens choose $a = 0$, denoted by $\bar{u}_0$, to zero. This normalization is without loss because $\bar{u}_0$ enters as a constant term independent of the persuasion thresholds and the cross-sectional distribution. The parameters w_h and w_l measure her expected benefit from securing support after the favorable and unfavorable realizations, respectively: w_h is larger when the favorable realization can be broadcast to a wide audience and converted into visible support, while w_l is larger when suppressing dissent after the unfavorable realization prevents coordination, elite defection, or costly opposition.

The dictator observes citizens' priors only through coarse summaries. She knows the average prior $\hat{\theta}$ and a dispersion lower bound $\underline{\sigma}^2$. The mean is inferred from aggregate behavior, such as turnout at regime rallies, participation in street protests, or compliance with local directives; the dispersion bound comes from evidence of polarization, such as disagreement in social-media posts.¹⁶ As preference falsification is pervasive, she seeks a payoff guarantee robust to small perturbations of these parameters.

Let $0 < \hat{\theta} < 1$ and $\underline{\sigma}^2 \in (0, \hat{\theta}(1 - \hat{\theta}))$, and define the ambiguity set

$$\Pi(\hat{\theta}, \underline{\sigma}^2) := \{F \in \Delta([0, 1]) : \mathbb{E}_F[\theta] = \hat{\theta}, \text{Var}_F(\theta) \geq \underline{\sigma}^2\}.$$

Because the dispersion lower bound rules out degenerate distributions, Nature cannot satisfy the moment restrictions by placing all citizens at the same belief. The inner problem is governed by the way the mean equality and the variance lower bound restrict Nature's ability to place citizens at or below the persuasion threshold. This threshold geometry generates the central comparative static: a higher average prior $\hat{\theta}$ need not improve the dictator's optimal value. Proposition 7 characterizes when this happens and why.

Boundary menu. In threshold coordinates, write $M_{h\text{-bd}}$ for the boundary menu with threshold pair $(0, 1/2)$ and $M_{l\text{-bd}}$ for the boundary menu with threshold pair $(1/2, 0)$. Under $M_{h\text{-bd}}$, every citizen contributes w_h , and citizens with $\theta > 1/2$ contribute an additional w_l ; under $M_{l\text{-bd}}$, the roles of w_h and w_l are reversed. Equivalently, the better boundary

¹⁶The lower-bound form captures a minimum degree of dispersion in citizens' priors. By contrast, an upper bound on variance would remain consistent with a nearly homogeneous population.

menu earns $\max\{w_h, w_l\}$ from every citizen and an additional $\min\{w_h, w_l\}$ from citizens with $\theta > 1/2$.

The two boundary menus, $(0, 1/2)$ and $(1/2, 0)$, are symmetric. Each pairs a narrative that makes one signal realization decisive evidence of strength with a narrative that makes the other uninformative. Thus one realization is fully insured, while the other is protected only by the moment-forced mass above $1/2$.

In the underlying narrative parameters, $(0, 0)$ makes the unfavorable realization l uninformative, while $(1, 1)$ makes the favorable realization h uninformative. The narrative $(1, 0)$ says that favorable news reveals strength, as when policy success reflects the dictator's competence; $(0, 1)$ says that unfavorable news reveals strength, as when enemies sabotage the dictator because she is competent.

Moment-forced tail mass under a boundary menu. The key object is the moment-forced tail mass above a persuasion threshold t . Write $\underline{T}(t)$ for the minimum, over all distributions satisfying the mean equality and variance lower bound, of the share of citizens with $\theta > t$. This is the least share of citizens the dictator can guarantee above the threshold after the relevant realization; its complement is the largest share Nature can keep at or below the margin. This threshold calculation is the specialization of Proposition 2 to $g(\theta) = (\theta, \theta^2)$ and payoff $\mathbf{1}\{\theta > t\}$: the dual is a quadratic minorant with contact in $\{0, t, 1\}$, and the worst case has two or three support points there.

Proposition 4 (Moment-forced tail mass under a boundary menu)

(i) Minimum tail mass. For every $\hat{\theta} \in (0, 1)$, $\underline{\sigma}^2 \in (0, \hat{\theta}(1 - \hat{\theta}))$, and threshold $t \in (0, 1)$, write

$$\underline{T}(t) := \inf_{F \in \Pi(\hat{\theta}, \underline{\sigma}^2)} F((t, 1])$$

for the minimum mass above t over the ambiguity set. Then

$$\underline{T}(t) = \max \left\{ \frac{\max\{\hat{\theta} - t, 0\}}{1 - t}, \frac{\underline{\sigma}^2 - \hat{\theta}(t - \hat{\theta})}{1 - t}, 0 \right\}. \quad (14)$$

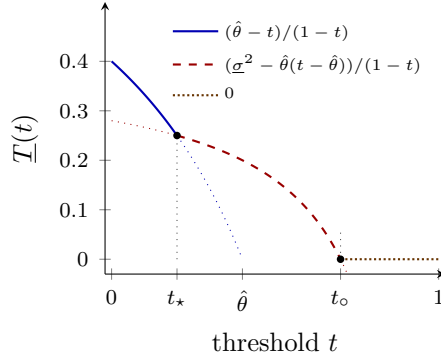
The infimum is attained by a two- or three-point distribution on $\{0, t, 1\}$, matching the finite-support contact pattern in Proposition 2.

(ii) Boundary-menu value at $\tau = 1/2$. Applying part (i) at $t = 1/2$, the payoff guarantees of the two boundary menus are

$$\min_{F \in \Pi(\hat{\theta}, \underline{\sigma}^2)} \int \tilde{v}_{(0, 1/2)} dF = w_h + w_l \cdot \underline{T}(1/2), \quad \min_{F \in \Pi(\hat{\theta}, \underline{\sigma}^2)} \int \tilde{v}_{(1/2, 0)} dF = w_l + w_h \cdot \underline{T}(1/2).$$

When $w_h \geq w_l$, the menu $(0, 1/2)$ weakly dominates and the relevant worst-case boundary value is $w_h + w_l \underline{T}(1/2)$; when $w_h < w_l$, $(1/2, 0)$ dominates with value $w_l + w_h \underline{T}(1/2)$. The dominance is strict whenever $w_h \neq w_l$, since $\underline{T}(1/2) < 1$.

Nature places mass at the lowest prior, at the persuasion threshold, and at the highest prior: mass at t absorbs moment requirements without entering the upper tail $F((t, 1])$, while mass at 1 appears only when the mean or variance bound forces mass above the threshold.



Worst-case tail mass $\underline{T}(t)$ at $\hat{\theta} = 0.4$, $\underline{\sigma}^2 = 0.12$

Figure 4: Worst-case tail mass $\underline{T}(t)$ for $\hat{\theta} = 0.4$ and $\underline{\sigma}^2 = 0.12$. The boundary application later evaluates this curve at $t = \tau = 1/2$.

The two breakpoints at which the active piece changes are

$$t_{\star} := \hat{\theta} - \frac{\underline{\sigma}^2}{1 - \hat{\theta}}, \quad t_o := \hat{\theta} + \frac{\underline{\sigma}^2}{\hat{\theta}};$$

the feasibility constraint $\underline{\sigma}^2 \in (0, \hat{\theta}(1 - \hat{\theta}))$ ensures $0 < t_{\star} < \hat{\theta} < t_o < 1$.

Three parameter regions determine which piece is active (Figure 4): the tail mass is zero, mean-forced, or variance-forced. See [Appendix B.1](#) for details.

Proposition 5 (Boundary optimality)

For every $w_h > 0$, $w_l > 0$, every $\hat{\theta} \in (0, 1)$, and every $\underline{\sigma}^2 \in (0, \hat{\theta}(1 - \hat{\theta}))$, a boundary menu is optimal. If $w_h \geq w_l$, the optimal boundary menu is $(0, 1/2)$ with value $w_h + w_l \underline{T}(1/2)$; if $w_h < w_l$, it is $(1/2, 0)$ with value $w_l + w_h \underline{T}(1/2)$. In the knife-edge case $w_h = w_l$, both boundary menus are optimal.

Thus, in the natural case $\tau = 1/2$, the optimal menu is a boundary menu. The realization with the larger payoff weight receives the zero threshold: h if $w_h \geq w_l$, and l otherwise.

The case $w_l > w_h$ is relevant for crisis-prone authoritarian politics. A larger w_l captures situations where securing support after an unfavorable realization is more valuable: unfavorable news can threaten the regime by revealing incompetence, enabling citizens to infer regime vulnerability, and encouraging elite defection. Literature on authoritarian survival emphasizes these asymmetric dangers: information manipulation is valuable because it can forestall coordinated uprisings ([Edmond, 2013](#)); censorship tends to target collective-action potential rather than criticism as such ([King et al., 2013](#)); preference falsification can generate sudden protest cascades once adverse events become common knowledge ([Kuran, 1991](#)); and modern “informational autocrats” survive by sustaining a competence image that unfavorable news threatens ([Guriev and Treisman, 2019](#); [Rozenas and Stukal, 2019](#)). Proposition 5 then selects the boundary menu $(1/2, 0)$. This corresponds to, for instance, a common authoritarian narrative in which protests, unrest, or policy failures are framed as sabotage by domestic opponents backed by foreign enemies. It resonates with the alternative-reality

account of propaganda in [Szeidl and Szucs \(2025\)](#), in which criticism or adverse evidence validates a populist narrative where attacks from hostile elites serve as proof that the politician is effectively challenging the corrupt establishment on behalf of the people.

A tighter variance lower bound shrinks $\Pi(\hat{\theta}, \underline{\sigma}^2)$ and restricts Nature's ability to remove citizens from persuaded tails. This yields the following monotonicity result.

Proposition 6 (Monotonicity in the variance lower bound)

(i) Weak monotonicity. *For any dictator primitives $(w_h, w_l, \hat{\theta})$ and any two feasible variance lower bounds $0 < \underline{\sigma}_1^2 \leq \underline{\sigma}_2^2 < \hat{\theta}(1 - \hat{\theta})$, the payoff guarantee of every menu is weakly higher under the tighter constraint:*

$$\min_{F \in \Pi(\hat{\theta}, \underline{\sigma}_1^2)} \int \tilde{v} dF \leq \min_{F \in \Pi(\hat{\theta}, \underline{\sigma}_2^2)} \int \tilde{v} dF \quad \forall M.$$

Consequently, the optimal value $\max_M \min_{F \in \Pi(\hat{\theta}, \underline{\sigma}^2)} \int \tilde{v} dF$ is weakly increasing in $\underline{\sigma}^2$.

(ii) Strict monotonicity. *On the variance-binding region for $T(1/2)$, the optimal value satisfies*

$$\max_M \min_{F \in \Pi(\hat{\theta}, \underline{\sigma}^2)} \int \tilde{v} dF = \max\{w_h, w_l\} + 2 \min\{w_h, w_l\}(\underline{\sigma}^2 - \hat{\theta}(1/2 - \hat{\theta})),$$

and on each connected component of this region the optimal value is strictly increasing in $\underline{\sigma}^2$ with derivative $2 \min\{w_h, w_l\} > 0$.

Greater observed dispersion among citizens' priors can raise the dictator's optimal value even though she seeks uniform political support: evidence of high enough polarization can rule out the least favorable population in which receiver mass is concentrated at the persuasion threshold and contributes no upper-tail payoff.

In [Gitmez and Molavi \(2022\)](#)'s Bayesian persuasion setting, dispersion is a property of the known population: a more diverse society makes public persuasion harder because the sender must appeal to more skeptical receivers without losing support elsewhere. Here, dispersion enters as a restriction on Nature's feasible set: a higher lower bound on prior dispersion shrinks $\Pi(\hat{\theta}, \underline{\sigma}^2)$. By Proposition 6, this weakly raises the optimal value.

The mean restriction has a subtler effect. It can raise the average prior by moving mass from lower-prior types toward the persuasion threshold, yet this shift can lower the worst-case mass strictly above the threshold.

Proposition 7 (Non-monotonicity in $\hat{\theta}$)

Fix $w_h > 0$, $w_l > 0$, and $\underline{\sigma}^2 \in (0, 1/4)$, and vary $\hat{\theta}$ over $\{\hat{\theta} \in (0, 1) : \hat{\theta}(1 - \hat{\theta}) > \underline{\sigma}^2\}$. At $\tau = 1/2$, the optimal value is

$$\begin{aligned} \max_M \min_{F \in \Pi(\hat{\theta}, \underline{\sigma}^2)} \int \tilde{v} dF &= \max\{w_h, w_l\} \\ &+ \min\{w_h, w_l\} \max\{2 \max\{\hat{\theta} - 1/2, 0\}, 2(\underline{\sigma}^2 - \hat{\theta}(1/2 - \hat{\theta})), 0\}. \end{aligned}$$

It fails to be monotone in $\hat{\theta}$ if and only if $\underline{\sigma}^2 < 3/16$. Its minimum equals $\max\{w_h, w_l\}$ whenever $\underline{\sigma}^2 \leq 1/16$, and equals $\max\{w_h, w_l\} + \min\{w_h, w_l\}(2\underline{\sigma}^2 - 1/8)$ at $\hat{\theta} = 1/4$ whenever $1/16 < \underline{\sigma}^2 < 3/16$. When $\underline{\sigma}^2 \geq 3/16$, the optimal value is strictly increasing in $\hat{\theta}$.

In the variance-binding region, Nature can raise the average prior by moving mass to $1/2$, where citizens are more favorable but still outside the strict upper tail. This lowers the worst-case upper-tail mass. Here $2(\underline{\sigma}^2 - \hat{\theta}(1/2 - \hat{\theta}))$ determines $\underline{T}(1/2)$; once the slack below the threshold is exhausted, further increases in $\hat{\theta}$ require more mass above $1/2$. See Figure 5.

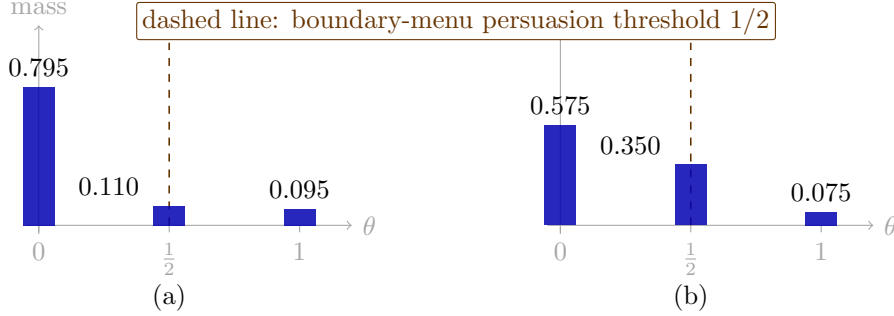


Figure 5: Worst-case distributions with $\underline{\sigma}^2 = 0.10$ and $\tau = 1/2$. Raising the mean from $\hat{\theta} = 0.15$ to $\hat{\theta} = 0.25$ lowers worst-case mass above $1/2$ from 9.5% to 7.5%.

This nonmonotonic pattern is possible only when the variance lower bound is not too high. With a high lower bound, Nature must keep enough mass far from the threshold throughout the feasible range of $\hat{\theta}$; increasing $\hat{\theta}$ then requires more mass on the high-prior side, and the optimal value is increasing. Different restrictions determine worst-case tail mass in different parameter regions: within the variance-binding region, a higher mean can lower the optimal value; within the mean-binding region, it uniformly helps.

By affecting the dictator's optimal value through the minimum tail mass above the persuasion threshold, ambiguity can also reshape the cross-realization allocation of persuasion.

Robust concentration versus benchmark spread. The boundary-menu result means that ambiguity leads the dictator to concentrate persuasion on the realization with the larger payoff weight, relying on the moment restrictions to force some citizens above the unfavorable-realization threshold. Whether this allocation differs from the benchmark problem at the worst-case distribution F^* depends on a saddle-point condition in [Appendix B.5](#). When it holds, $M_{h\text{-bd}}$ is benchmark-optimal at F^* if $w_h \geq w_l$, and $M_{l\text{-bd}}$ is benchmark-optimal at F^* if $w_l > w_h$. When it fails, a benchmark optimum at F^* can be chosen in the frontier interior, and ambiguity then changes both the optimal value and the allocation of persuasion across realizations.

Beyond $\tau = 1/2$. The same threshold geometry extends to arbitrary action thresholds. Formula (14) applies to any $\tau \in (0, 1)$, and a boundary menu with threshold τ has value

$$\max\{w_h, w_l\} + \min\{w_h, w_l\} \underline{T}(\tau)$$

whenever that boundary menu is optimal. On the variance-binding piece,

$$\underline{T}(\tau) = \frac{\underline{\sigma}^2 - \hat{\theta}(\tau - \hat{\theta})}{1 - \tau},$$

so the boundary value has $\hat{\theta}$ -derivative

$$\frac{\min\{w_h, w_l\}(2\hat{\theta} - \tau)}{1 - \tau}.$$

Thus the point at which the mean effect switches sign is $\hat{\theta} = \tau/2$; the turning point $1/4$ in Proposition 7 is the specialization to $\tau = 1/2$.

The negative mean effect carries over to the actual robust value. If $\hat{\theta} < \tau/2$, then $\hat{\theta} < 1/2$, so the primitive boundary-optimality condition in Appendix B.6 is automatically satisfied for the boundary menu associated with the larger payoff weight. Hence, whenever the variance piece is active below $\tau/2$, raising the average prior lowers the actual robust value. For a fixed variance lower bound, this decreasing low-mean interval is nonempty when

$$\underline{\sigma}^2 < \frac{\tau(2 - \tau)}{4}.$$

Outside the boundary-optimal regions, however, frontier-interior menus can dominate; the general- τ value then requires a case-by-case comparison along the persuasion frontier. The main text therefore keeps the symmetric threshold $\tau = 1/2$, where boundary optimality is global and the comparative statics apply throughout.

5 Conclusion

Robust narrative persuasion broadens the scope of persuasion. The signal space and the data-generating model may be outside the sender's control, governed by forces such as physical processes, other actors, or institutional rules. Information design is then unavailable. When interpretation remains malleable, however, narratives still allow the sender to pursue robust persuasion. The sender does not know the full distribution of priors, so she cannot solve a distribution-specific narrative persuasion problem. When she observes only a point estimate or confidence region for moments of that distribution, the sender can instead formulate the narrative persuasion problem over the associated ambiguity set. Moreover, under the paper's formulation, small errors in the moment information used to construct the ambiguity set do not make the resulting payoff guarantee drop discontinuously.

The analysis yields restrictions on behavior across signal realizations. If receivers take the sender-preferred action whenever they regard its aligned state as more likely, the optimal menu frames the realization with the higher sender payoff weight as diagnostic of the sender-preferred state. Moreover, for any strictly positive action threshold, no menu can make every receiver take the sender-preferred action after both signal realizations; if nearly all receivers do so after all events, forces outside narrative selection are likely at work. Each narrative covers the entire signal space, so adopting it shifts a receiver's belief after every realization; because public menus are chosen before the realization is known, persuading more receivers after one event comes at the cost of persuading fewer after the other. Without such cross-event consistency, ex post explanations can be more flexible and event-specific. The optimal menu also uses a limited narrative family. In persistent campaigns, similar narratives, for instance opponent sabotage, explain similar events. These restrictions illustrate the behavioral implications of robust narrative persuasion.

A Proofs for the results

A.1 Preliminaries

Definition A.1 (Lower semi-continuous envelope)

For a bounded function $f: \Theta \rightarrow \mathbb{R}$, the lower semi-continuous envelope of f is

$$\text{lsc}(f)(\theta) := \liminf_{\theta' \rightarrow \theta} f(\theta'), \quad \theta \in \Theta.$$

Equivalently, $\text{lsc}(f)$ is the largest lower semi-continuous function that is pointwise no greater than f .

Lemma A.1 (Properties of narrative selection)

Fix a realization $s \in S$. For any narrative m , write

$$\eta_s(m) := m(s \mid H) - m(s \mid L).$$

Take two distinct narratives m_i and m_j . If $\eta_s(m_i) = \eta_s(m_j)$, then their fit difference after s is constant in θ . If $\eta_s(m_i) < \eta_s(m_j)$, then

$$\text{Fit}(s, m_j, \theta) - \text{Fit}(s, m_i, \theta) = (m_j(s \mid L) - m_i(s \mid L)) + (\eta_s(m_j) - \eta_s(m_i))\theta$$

is strictly increasing in θ . Hence two fit functions are either identical after s or cross at most once after s . Whenever the fit ranking changes as type increases, it changes from the narrative with the lower value of η_s to the narrative with the higher value. Along the upper envelope of fit functions after s , the selected value of $\eta_s(m)$ is non-decreasing in type.

Proof of Lemma A.1. Direct subtraction from

$$\text{Fit}(s, m, \theta) = m(s \mid L) + \eta_s(m)\theta$$

gives the difference. Equal slopes make the difference constant; if that constant is zero, the two fit functions are identical after s , and otherwise their ranking never changes. If $\eta_s(m_i) < \eta_s(m_j)$, the difference has strictly positive slope, so the pair crosses at most once and only from m_i to m_j as θ increases.

To verify monotone active slopes on the upper envelope, take $\theta_1 < \theta_2$, and let m_1 and m_2 be fit maximizers at θ_1 and θ_2 , respectively. If $\eta_s(m_1) > \eta_s(m_2)$, then $D(\theta) := \text{Fit}(s, m_1, \theta) - \text{Fit}(s, m_2, \theta)$ is strictly increasing. Since m_1 is fit-maximal at θ_1 , $D(\theta_1) \geq 0$, so $D(\theta_2) > 0$, contradicting fit maximality of m_2 at θ_2 . Hence every active slope at θ_1 is weakly below every active slope at θ_2 . $\square$

Corollary A.1 (Monotone posteriors and action profiles)

For any menu $M \in \mathcal{M}$, the posterior $\hat{\mu}_s(\theta) := \mu_s^{m^*(s, \theta)}(\theta)$ is non-decreasing in θ for each $s \in S$, under the zero-fit completion in Definition A.2. Consequently, the induced action $a_\theta(s)$ is non-decreasing in θ for each $s \in S$.

Proof of Corollary A.1. Fix $s \in S$. If every narrative in M is s -null in the sense of Definition A.2, then the completed posterior after s is constant and the result is immediate. Otherwise the maximum fit after s is positive on $(0, 1)$.

For a fixed narrative m , the posterior

$$\mu_s^m(\theta) = \frac{\theta m(s | H)}{(1 - \theta)m(s | L) + \theta m(s | H)}$$

is weakly increasing in θ wherever the denominator is positive. Between switching points the selected narrative is fixed, so the selected posterior is weakly increasing there.

At an interior switching point θ^* from m_a to m_b as θ increases, Lemma A.1 gives $\eta_s(m_b) \geq \eta_s(m_a)$. If equality holds, fit equality implies that the two narratives have the same likelihood pair after s , so the posterior is unchanged at the switch. If $\eta_s(m_b) > \eta_s(m_a)$, fit equality at θ^* gives

$$m_b(s | L) - m_a(s | L) = -(\eta_s(m_b) - \eta_s(m_a))\theta^*$$

and hence

$$m_b(s | H) - m_a(s | H) = (\eta_s(m_b) - \eta_s(m_a))(1 - \theta^*) \geq 0.$$

Since the two narratives have the same positive fit at θ^* , the posterior under m_b is weakly higher than the posterior under m_a at the switch, with a strict increase whenever $\theta^* \in (0, 1)$ and the slopes differ. Sender-favorable tie-breaking selects the upward value at switches. Boundary zero-fit cases are completed by one-sided limits in Definition A.2. Therefore $\hat{\mu}_s$ is non-decreasing, and $a_\theta(s) = \mathbf{1}\{\hat{\mu}_s(\theta) \geq \tau\}$ is non-decreasing as well. $\square$

Lemma A.2 (Threshold formula)

Fix $s \in S$. If narrative m is adopted after s , whenever the posterior is well-defined,

$$\mu_s^m(\theta) \geq \tau \quad \text{iff} \quad \theta \geq \frac{\tau m(s | L)}{(1 - \tau)m(s | H) + \tau m(s | L)}.$$

This threshold is continuous in the likelihood pair $(m(s | H), m(s | L))$ wherever the denominator is positive.

Proof of Lemma A.2. The inequality

$$\frac{\theta m(s | H)}{\theta m(s | H) + (1 - \theta)m(s | L)} \geq \tau$$

rearranges to the threshold formula whenever the denominator is positive. Continuity is immediate on that domain. $\square$

Definition A.2 (Zero-fit completion)

Fix $s \in S$. A narrative is s -null if $m(s | H) = m(s | L) = 0$. If every narrative in a menu is s -null, then the realization has zero fit for every type under every offered narrative; I complete the posterior after s by $\hat{\mu}_s(\theta) = 0$ for all θ , so the induced persuasion threshold is $\theta_s = 1^+$, using the no-persuasion threshold convention of Definition 2.

If the menu contains at least one narrative that is not s -null, then the maximum fit after s is positive for every interior type. A zero-fit selected narrative can arise only at a boundary

belief $\theta \in \{0, 1\}$. At such a boundary, nearby interior types have positive fit under the selected upper-envelope segment, and Bayes' rule has a one-sided limit along those types. I extend (2) to the boundary belief by that limit. This completion preserves Lemmas A.1 and A.2 and Corollary A.1 and leaves the per-type payoff representation unchanged.

Lemma A.3 (Pointwise threshold monotonicity)

Extend the usual order on $[0, 1]$ by placing every real threshold weakly below the artificial threshold 1^+ . For any two threshold profiles $(\theta_s)_{s \in S}, (\theta'_s)_{s \in S} \in ([0, 1] \cup \{1^+\})^{|S|}$ with $\theta'_s \leq \theta_s$ for every $s \in S$, and any $\theta \in [0, 1]$,

$$v(\theta; (\theta'_s)_{s \in S}) \geq v(\theta; (\theta_s)_{s \in S}) \quad \text{and} \quad \tilde{v}(\theta; (\theta'_s)_{s \in S}) \geq \tilde{v}(\theta; (\theta_s)_{s \in S}).$$

Hence $\int \tilde{v}(\theta; (\theta'_s)_{s \in S}) F(d\theta) \geq \int \tilde{v}(\theta; (\theta_s)_{s \in S}) F(d\theta)$ for every $F \in \Delta([0, 1])$, and taking the infimum over any common non-empty subset of $\Delta([0, 1])$ preserves the inequality.

Proof of Lemma A.3. For v , $v(\theta; (\theta_s)_{s \in S}) = \bar{u}_0 + \sum_{s \in S} w_s \mathbf{1}\{\theta \geq \theta_s\}$ with $w_s > 0$, so lowering any threshold weakly enlarges its super-level set, giving the pointwise inequality for v .

For $\tilde{v}$, applying Definition A.1 to each indicator gives

$$\tilde{v}(\theta; (\theta_s)_{s \in S}) = \bar{u}_0 + \sum_{s \in S} w_s \tilde{I}_{\theta_s}(\theta),$$

where

$$\tilde{I}_t(\theta) := \begin{cases} \mathbf{1}\{\theta > t\} + \mathbf{1}\{\theta = 0\}, & t = 0, \\ \mathbf{1}\{\theta > t\}, & t \in (0, 1], \\ 0, & t = 1^+. \end{cases}$$

The function $\tilde{I}_{t'}$ weakly dominates $\tilde{I}_t$ pointwise whenever $t' \leq t$ in the extended order. This gives the pointwise inequality for $\tilde{v}$. Integration and infimum over a fixed feasible set preserve pointwise order. $\square$

Lemma A.4 (Compactness and attainment for moment ambiguity sets)

Let Θ be a compact metric space, $g: \Theta \rightarrow \mathbb{R}^d$ continuous, and $Y \subseteq \mathbb{R}^d$ closed. Set $\Pi := \{F \in \Delta(\Theta) : \mathbb{E}_F[g(\theta)] \in Y\}$ and assume $\Pi \neq \emptyset$. Then:

- (a) Π is weakly compact in $\Delta(\Theta)$.
- (b) For every bounded lower semi-continuous $f: \Theta \rightarrow \mathbb{R}$, $F \mapsto \int_{\Theta} f dF$ is weakly lower semi-continuous on $\Delta(\Theta)$.
- (c) $\inf_{F \in \Pi} \int_{\Theta} f dF$ is attained whenever f is bounded and lower semi-continuous.

Proof of Lemma A.4. $\Delta(\Theta)$ is weakly compact by Aliprantis and Border (2006, Theorem 15.11). Continuity of g on the compact Θ makes $F \mapsto \mathbb{E}_F[g(\theta)]$ weakly continuous, so Π is the pre-image of a closed set under a weakly continuous map, hence weakly closed; a weakly closed subset of a weakly compact set is weakly compact, giving (a). Part (b) is Aliprantis and Border (2006, Theorem 15.5). (c) follows from the Weierstrass theorem: a weakly lower semi-continuous function on a non-empty weakly compact set attains its infimum. $\square$

A.2 Proof of Theorem 1

Proof. For each $s \in S$, if no type is persuaded after s under M , choose an arbitrary narrative from M ; otherwise let $\theta_s(M)$ be the persuasion threshold and retain a narrative fit-maximal for type $\theta_s(M)$ after s and inducing a posterior of at least τ there. Let $M' \subseteq M$ be the set of retained narratives, so $|M'| \leq |S|$.

Fix s . If no type is persuaded under M , there is nothing to prove. Otherwise the retained narrative remains fit-maximal for $\theta_s(M)$ within M' because $M' \subseteq M$, so sender-favorable tie-breaking gives a posterior of at least τ for type $\theta_s(M)$ under M' . Monotonicity of the posterior then extends persuasion under M' to every $\theta \geq \theta_s(M)$, which is the realization- s upper interval of types induced to take action 1 under M .

Summing the per-realization inclusions over s and using $w_s > 0$ gives $v(\theta; M') \geq v(\theta; M)$ for every $\theta \in [0, 1]$; the infimum over $F \in \Pi$ gives $V(M') \geq V(M)$. $\square$

A.3 Proof of Theorem 2

Proof. Part (i): sufficiency. For each $s \in S$, set $m_s(s \mid H) = 1$, $m_s(s' \mid H) = 0$ for $s' \neq s$, and

$$m_s(s \mid L) = \frac{\theta_s(1 - \tau)}{(1 - \theta_s)\tau}.$$

Choose the remaining L -row entries with $0 \leq m_s(s' \mid L) \leq \min\{1, \theta_{s'}/(\tau(1 - \theta_{s'}))\}$ summing to $1 - m_s(s \mid L)$. Such a row exists because (7) gives the required slack:

$$1 - \frac{\theta_s(1 - \tau)}{(1 - \theta_s)\tau} \leq \sum_{s' \neq s} \min\left\{1, \frac{\theta_{s'}}{\tau(1 - \theta_{s'})}\right\}.$$

At signal s , m_s reaches the action threshold at θ_s (with $\theta_s = 0$ read in the right limit), while every $m_{s'}$ with $s' \neq s$ has fit $(1 - \theta_s)m_{s'}(s \mid L) \leq \theta_{s'}/\tau$. Hence m_s is fit-maximal at (s, θ_s) , and the induced threshold is θ_s .

Necessity. Let m_s be selected at (s, θ_s) in some implementing menu. The posterior condition gives

$$m_s(s \mid L) \leq \frac{\theta_s(1 - \tau)}{(1 - \theta_s)\tau} m_s(s \mid H) \leq \frac{\theta_s(1 - \tau)}{(1 - \theta_s)\tau}.$$

Fix $j \in S$. For $s \neq j$, fit-maximality of m_s at (s, θ_s) gives

$$\theta_s m_j(s \mid H) + (1 - \theta_s) m_j(s \mid L) \leq \theta_s m_s(s \mid H) + (1 - \theta_s) m_s(s \mid L) \leq \theta_s/\tau,$$

so $m_j(s \mid L) \leq \theta_s/(\tau(1 - \theta_s))$. Summing the L -row of m_j ,

$$1 \leq \frac{\theta_j(1 - \tau)}{(1 - \theta_j)\tau} + \sum_{s \neq j} \frac{\theta_s}{\tau(1 - \theta_s)}.$$

Multiplying by τ and maximizing over j yields (7).

Part (ii). Fix an ordering $r_{(1)} \leq \dots \leq r_{(|S|)}$. The maximum in (7) is then attained at $r_{(|S|)}$, so the binding inequality $\sum_{s \in S} r_s = \tau(1 + r_{(|S|)})$ rearranges to $\sum_{i=1}^{|S|-1} r_{(i)} + (1 - \tau) r_{(|S|)} = \tau$. The remaining $|S| - 1$ inequalities are weakly slack under this ordering. Permuting labels gives the global frontier. $\square$

Lemma A.5 (Upper-threshold dominance)

For any menu $M \in \mathcal{M}$, let $\bar{\theta}_s := \min\{\theta_s(M), \tau\}$ for each $s \in S$, with $\min\{1^+, \tau\} := \tau$. The profile $(\bar{\theta}_s)_{s \in S}$ is implementable, and

$$v(\theta; (\bar{\theta}_s)_{s \in S}) \geq v(\theta; M) \quad \text{and} \quad \tilde{v}(\theta; (\bar{\theta}_s)_{s \in S}) \geq \tilde{v}(\theta; M)$$

for every $\theta \in [0, 1]$.

Proof. If every $\theta_s(M)$ belongs to $[0, \tau]$, then $(\theta_s(M))_{s \in S} = (\bar{\theta}_s)_{s \in S}$ is already implementable and there is nothing to prove.

Otherwise fix $s^* \in S$ with $\bar{\theta}_{s^*} = \tau$, so $\bar{r}_{s^*} := \bar{\theta}_{s^*}/(1 - \bar{\theta}_{s^*}) = \tau/(1 - \tau)$. Since $\bar{\theta}_s \leq \tau$ for every $s \in S$ by construction, $\bar{r}_s \leq \tau/(1 - \tau)$ for every $s \in S$, so $\max_{s \in S} \bar{r}_s = \tau/(1 - \tau)$. Using $\bar{r}_s \geq 0$ for $s \neq s^*$,

$$\sum_{s \in S} \bar{r}_s \geq \bar{r}_{s^*} = \frac{\tau}{1 - \tau} = \tau \left(1 + \frac{\tau}{1 - \tau}\right) = \tau \left(1 + \max_{s \in S} \bar{r}_s\right).$$

This is (7) for $(\bar{\theta}_s)_{s \in S} \in [0, \tau]^{|S|}$, so the sufficiency direction of Theorem 2 gives a menu $\bar{M}$ implementing $(\bar{\theta}_s)_{s \in S}$.

Since $\bar{\theta}_s \leq \theta_s(M)$ for every $s \in S$ in the extended threshold order, Lemma A.3 gives $v(\theta; (\bar{\theta}_s)_{s \in S}) \geq v(\theta; (\theta_s(M))_{s \in S}) = v(\theta; M)$ for every $\theta \in [0, 1]$. The same argument gives $\tilde{v}(\theta; (\bar{\theta}_s)_{s \in S}) \geq \tilde{v}(\theta; M)$ for every $\theta \in [0, 1]$. $\square$

A.4 Proof of Corollary 2

Proof. Let $(\theta_s)_{s \in S}$ be any implementable profile, so $\theta_s \in [0, \tau]$ for every $s \in S$. By symmetry of (7), relabel the threshold coordinates so that lower thresholds are assigned to weakly larger payoff weights: after ordering $w_{s_1} \geq \dots \geq w_{s_n}$, take $0 \leq \theta_{s_1} \leq \dots \leq \theta_{s_n} \leq \tau$. This rearrangement weakly raises pointwise payoff by the standard pairwise swap argument: if $w_i \geq w_j$ but $\theta_i > \theta_j$, swapping the two thresholds changes payoff by $(w_i - w_j)\mathbf{1}\{\theta_j \leq \theta < \theta_i\} \geq 0$.

Write $r_{s_i} := \theta_{s_i}/(1 - \theta_{s_i})$. Since the ordered profile is implementable, Theorem 2 gives

$$L := \sum_{i=1}^{n-1} r_{s_i} + (1 - \tau)r_{s_n} \geq \tau.$$

If $L = \tau$, the profile is already on the persuasion frontier. If $L > \tau$, set $\lambda := \tau/L \in (0, 1)$ and define $r'_{s_i} := \lambda r_{s_i}$ for every i . The ordering is preserved, $r'_{s_i} \leq r_{s_i}$ for every i , and the new profile lies on the persuasion frontier:

$$\sum_{i=1}^{n-1} r'_{s_i} + (1 - \tau)r'_{s_n} = \tau.$$

Converting back by $\theta'_{s_i} = r'_{s_i}/(1 + r'_{s_i})$ gives an implementable profile by Theorem 2. Since every threshold weakly falls, Lemma A.3 gives pointwise weak dominance. The maximum can therefore be restricted to the persuasion frontier. $\square$

A.5 Proof of Proposition 1

Proof for Proposition 1. By Lemma A.5, every menu is pointwise weakly dominated in v by one inducing an implementable threshold profile in $[0, \tau]^{|S|}$, so the sender can restrict attention to that set without loss. In odds coordinates this set is closed by (7) (Theorem 2); hence it is compact. For each $F \in \Pi$, the map

$$(\theta_s)_{s \in S} \mapsto \int v(\theta; (\theta_s)_{s \in S}) F(d\theta)$$

is upper semi-continuous: for any convergent sequence of threshold profiles, apply Aliprantis and Border (2006, Lemma 11.20) to the bounded nonnegative functions obtained by subtracting the finite sum of upper semi-continuous indicators in (5) from $\bar{u}_0 + \sum_{s \in S} w_s$. Therefore V , as the infimum over $F \in \Pi$ of upper semi-continuous functions on a compact set, is upper semi-continuous (Aliprantis and Border, 2006, Lemma 2.41). The Weierstrass theorem gives an optimal threshold profile, and Theorem 2 gives a finite menu implementing it. $\square$

A.6 Default-model extension

This extension lets receivers choose from $M \cup \{\pi\}$, where π is the fixed data-generating model. The sender still chooses only M . For a menu M , let $\theta_s^\pi(M)$ denote the persuasion threshold after realization $s \in S$ under this enlarged choice set. Define

$$\theta_s^\pi := \frac{\tau \pi(s \mid L)}{1 - \tau \pi(s \mid H) + \tau \pi(s \mid L)}, \quad s \in S. \quad (\text{A.1})$$

Proposition A.1 (Reduction and existence under the default model)

Under the default model:

- (i) *for every menu M and every realization $s \in S$, the induced persuasion set is an upper set, hence is represented by a threshold $\theta_s^\pi(M)$;*
- (ii) *every menu is pointwise weakly dominated by a sub-menu of M with at most $|S|$ narratives;*
- (iii) *an optimal menu exists.*

Proof. Apply Lemma A.1 and Corollary A.1 to the enlarged menu $M \cup \{\pi\}$. This gives monotone selected posteriors and upper persuasion sets after every realization.

For each $s \in S$, if no type is persuaded after s , retain no narrative for that realization. If the threshold is set by a sender-provided narrative, retain one such narrative. If the threshold is set by π , retain no sender-provided narrative for that realization. If no narrative from M has been retained after this step, keep an arbitrary element of M . The retained menu has at most $|S|$ narratives. After adjoining the fixed default model π , the threshold-setting argument used in Theorem 1 applies realization by realization: the retained threshold-setting object, whether it is sender-provided or π , is still available, and its posterior is non-decreasing in type. Hence every type persuaded under $M \cup \{\pi\}$ remains persuaded under the retained sub-menu together with π .

For existence, Proposition A.2 below identifies the feasible threshold profiles under the default model as the intersection of the compact implementability set in Theorem 2 with the closed coordinatewise lower bounds in (A.1). The upper semi-continuity argument in the proof of Proposition 1 therefore applies on this compact feasible set, and Proposition A.2 supplies an implementing menu for an optimal profile. $\square$

Proposition A.2 (Default-model implementability)

A threshold profile $(\theta_s)_{s \in S} \in [0, \tau]^{|S|}$ is implementable under the default model if and only if it is implementable in the problem without the default model and

$$\theta_s \geq \underline{\theta}_s^\pi \quad \text{for every } s \in S. \quad (\text{A.2})$$

Consequently, whenever $\pi(s \mid L) > 0$ for some realization s , the default-model feasible set rules out profiles with $\theta_s < \underline{\theta}_s^\pi$.

Proof. Necessity. If $(\theta_s)_{s \in S}$ is induced by $M \cup \{\pi\}$, then it is implementable without the default model by the menu $M \cup \{\pi\}$, since π is itself a stochastic kernel from Ω to $\Delta(S)$ and sender menus are unrestricted.

Fix $s \in S$. If π is selected at the marginal type θ_s and persuades, then θ_s is at least the persuasion threshold of π after s :

$$\frac{\tau \pi(s \mid L)}{(1 - \tau)\pi(s \mid H) + \tau \pi(s \mid L)}.$$

This number is weakly above $\underline{\theta}_s^\pi$, because its denominator is weakly smaller than the denominator in (A.1). Hence (A.2) holds.

If a sender-provided narrative m_s is selected at the marginal type and persuades, then

$$\text{Fit}(s, m_s, \theta_s) \leq \frac{\theta_s}{\tau},$$

because posterior persuasion implies

$$\theta_s m_s(s \mid H) \geq \tau \text{Fit}(s, m_s, \theta_s)$$

and $m_s(s \mid H) \leq 1$. Since m_s must weakly beat the default model at the same type,

$$(1 - \theta_s)\pi(s \mid L) + \theta_s\pi(s \mid H) \leq \frac{\theta_s}{\tau}.$$

Rearranging gives $\theta_s \geq \underline{\theta}_s^\pi$.

Sufficiency. Take a profile that satisfies Theorem 2 and (A.2). Use the constructive menu from the proof of Theorem 2: for each $s \in S$, the s -threshold narrative m_s has $m_s(s \mid H) = 1$ and reaches posterior τ at θ_s . Its fit at (s, θ_s) is θ_s/τ . The bound (A.2) is

$$\frac{\theta_s}{\tau} \geq (1 - \theta_s)\pi(s \mid L) + \theta_s\pi(s \mid H),$$

so m_s weakly beats the default model at the marginal type.

For $\theta < \theta_s$, the same constructed narrative has fit strictly above θ/τ after realization s . If the default model persuaded such a type, then

$$\text{Fit}(s, \pi, \theta) \leq \frac{\theta \pi(s | H)}{\tau} \leq \frac{\theta}{\tau},$$

so the default model could not be selected against m_s . The sender-provided narratives already implement the threshold profile without the default model by Theorem 2; adding π therefore does not create persuasion below any θ_s .

It remains to check that adding π does not destroy persuasion above θ_s . The default model itself reaches posterior τ after s at

$$\alpha_s := \frac{\tau \pi(s | L)}{(1 - \tau)\pi(s | H) + \tau\pi(s | L)}$$

when the denominator is positive. If θ is weakly above this value, selection of π is harmless because π persuades. Consider lower values of θ in the interval starting at θ_s . The constructed narrative has fit

$$c_{\theta_s}(\theta) := \theta + (1 - \theta) \frac{\theta_s(1 - \tau)}{(1 - \theta_s)\tau},$$

while the default model has fit

$$d(\theta) := (1 - \theta)\pi(s | L) + \theta\pi(s | H).$$

Both are affine in θ . The lower bound (A.2) gives $c_{\theta_s}(\theta_s) \geq d(\theta_s)$. To check the other endpoint, write $p := \pi(s | H)$, $q := \pi(s | L)$, and

$$b_s := \frac{\tau q}{1 - \tau p + \tau q} = \frac{\theta_s^\pi}{1 - \tau p + \tau q}.$$

For fixed θ , $c_t(\theta)$ is weakly increasing in t . Hence $\theta_s \geq b_s$ gives $c_{\theta_s}(\alpha_s) \geq c_{b_s}(\alpha_s)$ whenever α_s is defined. A direct calculation gives

$$c_{b_s}(\alpha_s) - d(\alpha_s) = \frac{\tau q(1 - p)^2}{(1 - \tau p)((1 - \tau)p + \tau q)} \geq 0,$$

with zero-probability boundary cases obtained by continuity. Hence the constructed narrative weakly beats the default model at the default model's own persuasion threshold whenever (A.2) holds. Since $c_{\theta_s} - d$ is affine, it is nonnegative throughout the interval between θ_s and α_s . Thus the default model cannot displace the threshold narrative on the part of the upper interval where the default model would not persuade. Persuasion is preserved for every $\theta \geq \theta_s$, and the profile is implementable under the default model. $\square$

For any threshold profile implementable under the default model, the inner minimization problem is unchanged after restricting the feasible threshold profiles by (A.2). Hence the payoff-guarantee equivalence, dual characterization, finite-support worst-case distribution, and contact-set characterization apply on the restricted feasible set.

A.7 Beyond binary states

This subsection records the formal condition behind the scope discussion in Section 3.3.

Proposition A.3 (Scalar-statistic threshold sufficiency)

Let $\mathcal{Q}$ be a primitive state space, let $X = [\underline{x}, \bar{x}] \subset \mathbb{R}$ be an ordered scalar index space, and let $T : \Delta(\mathcal{Q}) \rightarrow X$ be a posterior statistic. For each menu M and realization s , let $Z_s^M(\nu)$ be the scalar posterior index generated by the narrative selected after s from M when the primitive posterior is $\nu \in \Delta(\mathcal{Q})$. Suppose:

(i) the induced posterior index factors through T : for every M and s , there exists a map $z_s^M : X \rightarrow X$ such that

$$Z_s^M(\nu) = z_s^M(T(\nu)) \quad \text{for every } \nu \in \Delta(\mathcal{Q});$$

(ii) z_s^M is non-decreasing for every M and s ;

(iii) the receiver takes action 1 if and only if $z_s^M(x) \geq \kappa$ for a fixed cutoff κ ;

(iv) for every M and s , the upper set $\{x : z_s^M(x) \geq \kappa\}$ contains its lower endpoint whenever it is non-empty;

(v) the sender's ordinal comparison of menus depends on primitive posterior beliefs only through the induced scalar-index thresholds;

(vi) for the sender payoff representation below, the sender's payoff from action 1 after realization s is $w_s > 0$, and residual features of the primitive posterior do not enter her payoff.

Then the primitive model is behavior-equivalent to a scalar-index model with receiver index $x = T(\nu)$: conditions (i)-(iv) give receiver behavior equivalence, and condition (v) gives sender menu-choice equivalence. For every menu M and realization s , the set $\{x : z_s^M(x) \geq \kappa\}$ is an upper interval containing its lower endpoint whenever non-empty. Let

$$\theta_s(M) := \inf\{x : z_s^M(x) \geq \kappa\},$$

with the convention that $\theta_s(M) = \bar{x}^+$ when the set is empty, where $\bar{x}^+$ is an artificial point above $\bar{x}$ and $\mathbf{1}\{x \geq \bar{x}^+\} = 0$ for every $x \in X$. If condition (vi) also holds, the sender's per-type payoff is

$$\bar{u}_0 + \sum_{s \in S} w_s \mathbf{1}\{x \geq \theta_s(M)\}.$$

Thus receiver behavior is represented by the threshold profile $(\theta_s(M))_{s \in S}$; under condition (v), the same profile represents sender menu choice; and under condition (vi), it also represents the sender's cardinal payoff-relevant part of the menu.

Proof. Condition (i) says that primitive posteriors with the same statistic T produce the same induced posterior index after every realization and menu. Hence the primitive posterior ν can be replaced, for receiver behavior purposes, by the scalar index $x = T(\nu)$. Condition (iii) then gives the action rule in the scalar-index model.

By conditions (ii) and (iv), for each M and s the super-level set $\{x : z_s^M(x) \geq \kappa\}$ is an upper interval that contains its lower endpoint whenever non-empty. Its lower endpoint is $\theta_s(M)$ under the stated convention, so the induced action after s is $\mathbf{1}\{x \geq \theta_s(M)\}$. This proves the receiver-side threshold representation. Condition (v) makes the scalar-index threshold profile sufficient for the sender's menu choice. Condition (vi) rules out residual payoff terms that distinguish primitive posteriors with the same scalar statistic. Under that additional payoff restriction, the sender's per-type payoff is the expression displayed in the proposition, and the menu is payoff-relevant only through $(\theta_s(M))_{s \in S}$. $\square$

Binary partition. Let $H \subseteq \mathcal{Q}$, write $L = \mathcal{Q} \setminus H$, and set $T(\nu) = \nu(H)$. If a seller wants the receiver to buy and the receiver buys whenever product quality q exceeds an adequacy cutoff q^* , the natural partition is $H = \{q \geq q^*\}$ and $L = \{q < q^*\}$. This partition is receiver-behavior-relevant when conditions (i)-(iv) in Proposition A.3 hold for $T(\nu) = \nu(H)$: action choice and the selected posterior index factor through the partition-induced coarsening, so receiver behavior is invariant to redistributions of posterior probability within H and within L . A primitive sufficient condition is that receiver utility differences and narrative likelihoods are measurable with respect to $\sigma(\{H, L\})$ and the resulting action upper set contains its lower endpoint whenever non-empty. Behavior equivalence including sender menu choice additionally requires condition (v). The per-type payoff representation additionally requires condition (vi) for the sender objective.

Scalar posterior statistic. If the primitive state is real-valued and the receiver takes action 1 whenever the posterior mean exceeds a cutoff, take $T(\nu) = \int q d\nu(q)$. The posterior mean then plays the same role as the belief on H , provided that the selected posterior index factors through T . Sender behavior equivalence additionally requires the sender's menu ordering to depend only on the scalar-index thresholds; payoff-guarantee equivalence further requires payoff increments to depend on ν only through $T(\nu)$. This is the scalar-statistic case discussed in Section 3.3. If any condition in Proposition A.3 fails, the scalar index need not summarize the design problem: the action rule may fail to be an upper interval, the relevant upper interval may fail to include its lower endpoint, the induced posterior index may fail to be monotone in the index, the sender's menu ordering may use residual posterior features, or the sender's payoff may use posterior features not captured by the thresholds.

A.8 Proof of Lemma 1

Proof of Lemma 1. Fix M and write $C := \text{conv } g([0, 1])$. Since $\tilde{v}(\cdot; M) \leq v(\cdot; M)$ pointwise,

$$\inf_{F \in \Pi} \widetilde{W}(M, F) \leq \inf_{F \in \Pi} W(M, F).$$

It remains to prove the reverse inequality.

Fix $F \in \Pi$ and let $y := \int g dF \in Y$. For $\delta > 0$, define the left-shift map $T_\delta(\theta) := \max\{0, \theta - \delta\}$ and let $Q_\delta := (T_\delta)_\# F$. For each positive real threshold t , $\mathbf{1}\{T_\delta(\theta) \geq t\} \rightarrow \mathbf{1}\{\theta > t\}$. For a zero threshold the indicator is always one, and for the artificial threshold 1^+ it is always zero. Hence $v(T_\delta(\theta); M) \rightarrow \tilde{v}(\theta; M)$ pointwise. Bounded convergence gives

$$\int v(\theta; M) Q_\delta(d\theta) \rightarrow \int \tilde{v}(\theta; M) F(d\theta).$$

Uniform continuity of g gives $q_\delta := \int g dQ_\delta \rightarrow y$.

Let $y_0 \in Y \cap \text{ri} C$ be a star center, and choose $H_0 \in \Delta([0, 1])$ with $\int g dH_0 = y_0$. Since y_0 is in the relative interior of C , there is $R > 0$ such that the relative ball $B(y_0, R)$ lies in C . Fix $\varepsilon \in (0, 1)$ and set $z_\varepsilon := (1 - \varepsilon)y + \varepsilon y_0 \in Y$. Convexity of C and the relative ball around y_0 imply $B(z_\varepsilon, \varepsilon R) \subset C$. Choose $\delta = \delta(\varepsilon)$ so that

$$\|q_\delta - y\| \leq \frac{\varepsilon^2 R}{(1 - \varepsilon)^2}.$$

Let $P_\varepsilon := (1 - \varepsilon)Q_\delta + \varepsilon H_0$, with moment $p_\varepsilon = (1 - \varepsilon)q_\delta + \varepsilon y_0$, and define

$$c_\varepsilon := p_\varepsilon + \frac{z_\varepsilon - p_\varepsilon}{\varepsilon}.$$

Then $\|c_\varepsilon - z_\varepsilon\| \leq \varepsilon R$, so $c_\varepsilon \in C$. Choose $H_\varepsilon \in \Delta([0, 1])$ with $\int g dH_\varepsilon = c_\varepsilon$, and set

$$G_\varepsilon := (1 - \varepsilon)P_\varepsilon + \varepsilon H_\varepsilon.$$

The moment of G_ε is $(1 - \varepsilon)p_\varepsilon + \varepsilon c_\varepsilon = z_\varepsilon \in Y$, so $G_\varepsilon \in \Pi$. Since v is bounded and the total correction mass vanishes with ε ,

$$\int v(\theta; M) G_\varepsilon(d\theta) \rightarrow \int \tilde{v}(\theta; M) F(d\theta).$$

Therefore $\inf_{G \in \Pi} W(M, G) \leq \widetilde{W}(M, F)$. Taking the infimum over $F \in \Pi$ proves the reverse inequality.

The payoff $\tilde{v}(\cdot, M)$ is bounded and lower semi-continuous in θ , so Lemma A.4 gives an attaining $F^* \in \Pi$. Hence $\widetilde{V}(M) = V(M)$ for every menu M , and the maximizers and maximum values of the two objective functions coincide by Proposition 1. $\square$

Proposition A.4 (Positive payoff guarantee above $\bar{u}_0$)

Suppose $\Pi = \mathcal{F}(g, Y)$, where $g : [0, 1] \rightarrow \mathbb{R}^d$ is continuous, $Y \subseteq \text{conv } g([0, 1])$ is closed, and $\Pi \neq \emptyset$. Fix an implementable threshold profile $(\theta_s)_{s \in S}$ and let

$$\Delta(\theta) := \tilde{v}(\theta; (\theta_s)_{s \in S}) - \bar{u}_0.$$

If $\theta_s = 0$ for some $s \in S$, then

$$\inf_{F \in \Pi} \int_0^1 \Delta(\theta) F(d\theta) > 0.$$

If $\theta_s > 0$ for every $s \in S$ and $\underline{\theta} := \min_{s \in S} \theta_s$, then

$$\inf_{F \in \Pi} \int_0^1 \Delta(\theta) F(d\theta) > 0 \iff Y \cap \text{conv } g([0, \underline{\theta}]) = \emptyset.$$

Proof. If $\theta_s = 0$ for some s , then the lower semi-continuous payoff convention in (13) gives

$$\mathbf{1}\{\theta > \theta_s\} + \mathbf{1}\{\theta_s = 0\}\mathbf{1}\{\theta = 0\} = 1$$

for every $\theta \in [0, 1]$. Since $w_s > 0$, $\Delta(\theta) \geq w_s$ pointwise, and the infimum is at least w_s .

Now suppose $\theta_s > 0$ for every s and write $\underline{\theta} = \min_s \theta_s$. Since all w_s are strictly positive,

$$\Delta(\theta) = \sum_{s \in S} w_s \mathbf{1}\{\theta > \theta_s\}$$

is strictly positive on $(\underline{\theta}, 1]$. Hence

$$\inf_{F \in \Pi} \int_0^1 \Delta(\theta) F(d\theta) > 0 \iff \inf_{F \in \Pi} F((\underline{\theta}, 1]) > 0.$$

It remains to classify the right-hand condition.

Let $A = (\underline{\theta}, 1]$. If $Y \cap \text{conv } g([0, \underline{\theta}]) \neq \emptyset$, choose $y \in Y \cap \text{conv } g([0, \underline{\theta}])$. By the definition of convex hull on the compact interval, there is $F \in \Delta([0, \underline{\theta}])$ with $\mathbb{E}_F[g(\theta)] = y$. Then $F \in \Pi$ and $F(A) = 0$, so $\inf_{G \in \Pi} G(A) = 0$.

Conversely, suppose $\inf_{F \in \Pi} F(A) = 0$. Choose $F_n \in \Pi$ with $F_n(A) \rightarrow 0$. Since $\Delta([0, 1])$ is weakly compact, a subsequence converges weakly to some $F^* \in \Delta([0, 1])$. The set A is open in the relative topology of $[0, 1]$, so Portmanteau's theorem gives

$$F^*(A) \leq \liminf_n F_n(A) = 0.$$

Thus F^* is supported on $[0, \underline{\theta}]$. Continuity of g gives

$$\mathbb{E}_{F_n}[g(\theta)] \rightarrow \mathbb{E}_{F^*}[g(\theta)]$$

along the convergent subsequence. Since each moment vector lies in the closed set Y , the limit belongs to Y . Since F^* is supported on $[0, \underline{\theta}]$, the same limit also belongs to $\text{conv } g([0, \underline{\theta}])$. Therefore $Y \cap \text{conv } g([0, \underline{\theta}]) \neq \emptyset$. $\square$

Public randomization. Let $\Delta_f(\mathcal{M})$ denote the finitely supported probability measures on the menu space. For $\nu \in \Delta_f(\mathcal{M})$, let $\widetilde{W}(\nu, F)$ be the finite ν -average of $\widetilde{W}(M, F)$. If the sender could publicly randomize over finitely many menus and the realized menu were observed before receivers act, the mixed problem for $\widetilde{W}$ would satisfy a minimax identity. Since Y is convex, $\Pi = \mathcal{F}(g, Y)$ is convex; by Lemma A.4(a), it is weakly compact.

Set $R := \inf_{F \in \Pi} \sup_{M \in \mathcal{M}} \widetilde{W}(M, F)$ and fix $a < R$. For each menu M , the set

$$U_M := \{F \in \Pi : \widetilde{W}(M, F) > a\}$$

is open in the weak topology because $F \mapsto \widetilde{W}(M, F)$ is lower semi-continuous. The family $\{U_M : M \in \mathcal{M}\}$ covers Π , so compactness gives menus $M_1, \dots, M_k$ with $\Pi \subseteq \bigcup_{i=1}^k U_{M_i}$. On the finite simplex Δ_k and the compact convex set Π , the payoff

$$(\nu, F) \mapsto \sum_{i=1}^k \nu_i \widetilde{W}(M_i, F)$$

is linear and continuous in ν , affine and lower semi-continuous in F . Sion's minimax theorem gives

$$\sup_{\nu \in \Delta_k} \inf_{F \in \Pi} \sum_{i=1}^k \nu_i \widetilde{W}(M_i, F) = \inf_{F \in \Pi} \max_{1 \leq i \leq k} \widetilde{W}(M_i, F) > a.$$

Letting $a \uparrow R$ and using the pointwise inequality $\inf_F \widetilde{W}(\nu, F) \leq \inf_F \sup_M \widetilde{W}(M, F)$ yields

$$\sup_{\nu \in \Delta_f(\mathcal{M})} \inf_{F \in \Pi} \widetilde{W}(\nu, F) = \inf_{F \in \Pi} \sup_{M \in \mathcal{M}} \widetilde{W}(M, F).$$

Thus public randomization over finite lotteries eliminates the max-min gap relative to the distribution-specific benchmark; deterministic public menus are the environment in which ambiguity matters.

A.9 Proof of Corollary 3

Proof of Corollary 3. By Lemmas 1 and A.5, every menu $M \in \mathcal{M}$ is pointwise weakly dominated in $\tilde{v}$ by a menu inducing an implementable threshold profile in $[0, \tau]^{|S|}$. Hence the outer maximum over $\mathcal{M}$ can be restricted to implementable threshold profiles. For $\tilde{v}$, the pairwise swap used in Corollary 2 remains valid: if $w_i \geq w_j$ and $\theta_i > \theta_j$, then swapping the two thresholds changes the lower semi-continuous payoff by

$$(w_i - w_j)(\tilde{I}_{\theta_j}(\theta) - \tilde{I}_{\theta_i}(\theta)) \geq 0$$

for every type θ , where $\tilde{I}_t$ is defined in the proof of Lemma A.3. After this ordering step, Corollary 2 scales the odds coordinates to a persuasion frontier profile and weakly lowers every threshold. By Lemma A.3, this raises $\tilde{v}$ pointwise in θ , and therefore weakly raises the robust payoff guarantee after taking the infimum over $F \in \Pi$. Combined with attainment in Proposition 1, this gives the max-min identity in Corollary 3.

When $S = \{h, l\}$ and $w_h \geq w_l$, the persuasion frontier is the h -first branch $\psi(\theta_h, \theta_l) = 0$ with $0 \leq \theta_h \leq \theta_l \leq \tau$. The case $w_l > w_h$ follows by exchanging h and l and replacing ψ by ϕ . $\square$

A.10 Proof of Proposition 2

Proof of Proposition 2. Set $f(\theta) := \tilde{v}(\theta; (\theta_s)_{s \in S})$ and $C := \text{conv } g([0, 1])$. By Definition A.1, f is bounded and lower semi-continuous on the compact interval. Primal attainment follows from Lemma A.4.

Consider

$$K := \{(x, z) \in C \times \mathbb{R} : \text{there is } F \in \Delta([0, 1]) \text{ with } x = \mathbb{E}_F g(\theta) \text{ and } z \geq \mathbb{E}_F f(\theta)\}.$$

The set K is convex, closed in the relative space $\text{aff } C \times \mathbb{R}$, and upward closed in its last coordinate. Closedness follows from weak compactness of $\Delta([0, 1])$, continuity of g , and lower semi-continuity of $F \mapsto \int f dF$. Let

$$p := \min_{F \in \Pi} \int f dF$$

and define $D := K - (Y \times \{0\})$. Then D is closed, convex, and upward closed, and

$$p = \min\{z : (0, z) \in D\}.$$

A supporting hyperplane to D at $(0, p)$ gives a nonzero vector (a, b) such that

$$a^\top u + bz \geq bp \quad \text{for every } (u, z) \in D.$$

Upward closedness implies $b \geq 0$. In fact $b > 0$: if $b = 0$, then $a^\top(x - y) \geq 0$ for every $x \in C$ and $y \in Y$. Taking a star center $y_0 \in Y \cap \text{ri } C$ and moving in both relative directions around y_0 inside C forces $a = 0$ on $\text{aff } C$, contradicting that the supporting vector is nonzero on the relative space. Normalize $b = 1$.

For every θ and every $y \in Y$, applying the support inequality to the degenerate distribution at θ gives

$$a^\top(g(\theta) - y) + f(\theta) \geq p.$$

Thus

$$f(\theta) \geq p + \sup_{y \in Y} a^\top y - a^\top g(\theta).$$

Set $\lambda^* := -a$ and $\gamma^* := p + \sup_{y \in Y} a^\top y$. Then $\gamma^* + (\lambda^*)^\top g(\theta) \leq f(\theta)$ for every θ , and

$$\gamma^* + \inf_{y \in Y} (\lambda^*)^\top y = p.$$

The dual value is therefore at least p , while every feasible affine minorant gives a lower bound on the primal value. Strong duality and dual existence follow.

Let F^* be a primal minimizer and $y^* := \mathbb{E}_{F^*} g(\theta)$. The inequalities

$$p = \int f dF^* \geq \gamma^* + (\lambda^*)^\top y^* \geq \gamma^* + \inf_{y \in Y} (\lambda^*)^\top y = p$$

all bind. Hence $y^* \in \arg \min_{y \in Y} (\lambda^*)^\top y$ and F^* puts mass only on the contact set. Finally, fix such a minimizing moment y^* . Among distributions satisfying $\mathbb{E}_F g = y^*$ and minimizing $\int f dF$, choose an extreme point of the minimizing face. By [Winkler \(1988, Theorem 2.1\)](#), it has support of size at most $d + 1$. When $g(\theta) = (\theta, \theta^2, \dots, \theta^d)$, the dual minorant is a polynomial of degree at most d . $\square$

A.11 Proof of Proposition 3

Proof of Proposition 3. The saddle inequalities $\widetilde{W}(M, F^*) \leq \widetilde{W}(M^*, F^*) \leq \widetilde{W}(M^*, F)$ for all $M \in \mathcal{M}$, $F \in \Pi$ give $M^* \in \arg \max_M \widetilde{W}(M, F^*)$ and $\widetilde{W}(M^*, F^*) = \min_F \widetilde{W}(M^*, F)$. Hence

$$\min_F \widetilde{W}(M, F) \leq \widetilde{W}(M, F^*) \leq \min_F \widetilde{W}(M^*, F) \quad \text{for every } M \in \mathcal{M},$$

so $M^* \in \arg \max_M \min_F \widetilde{W}(M, F)$. Lemma 1 replaces $\widetilde{W}$ by W inside the infimum, giving $M^* \in \arg \max_M \inf_F W(M, F)$. $\square$

A.12 Saddle-point criterion for benchmark selection

Proposition A.5 (Saddle-point criterion for benchmark selection)

Let M^* induce threshold profile $\theta^* = (\theta_s^*)_{s \in S}$, and fix $F^* \in \arg \min_{F \in \Pi} \widetilde{W}(M^*, F)$. Then

(M^*, F^*) is a saddle point of $\widetilde{W}$ on $\mathcal{M} \times \Pi$ if and only if

$$\int \tilde{v}(\theta; (\theta_s)_{s \in S}) F^*(d\theta) \leq \int \tilde{v}(\theta; \theta^*) F^*(d\theta) \quad (\text{A.3})$$

for every threshold profile $(\theta_s)_{s \in S}$ for which there exists an enumeration $s_1, \dots, s_n$ of S , $n = |S|$, satisfying

$$w_{s_1} \geq \dots \geq w_{s_n}, \quad 0 \leq \theta_{s_1} \leq \dots \leq \theta_{s_n} \leq \tau,$$

and

$$\sum_{i=1}^{n-1} \frac{\theta_{s_i}}{1 - \theta_{s_i}} + (1 - \tau) \frac{\theta_{s_n}}{1 - \theta_{s_n}} = \tau.$$

Whenever (A.3) holds, Proposition 3 implies that M^* is benchmark-optimal at F^* .

Proof of Proposition A.5. Since F^* minimizes $\widetilde{W}(M^*, \cdot)$ over Π , the right saddle inequality

$$\widetilde{W}(M^*, F^*) \leq \widetilde{W}(M^*, F) \quad \text{for every } F \in \Pi$$

already holds. It remains to characterize the left saddle inequality. By Lemma A.5, every menu is pointwise weakly dominated in $\tilde{v}$ by an implementable threshold profile in $[0, \tau]^{|S|}$. The pairwise swap calculation in the proof of Corollary 3 assigns weakly lower thresholds to weakly higher payoff weights without lowering $\tilde{v}$, and Corollary 2 and Lemma A.3 then move the ordered profile to the persuasion frontier without lowering $\tilde{v}$. Integrating against F^* preserves pointwise dominance. Hence

$$\widetilde{W}(M, F^*) \leq \widetilde{W}(M^*, F^*) \quad \text{for every } M \in \mathcal{M}$$

holds if and only if every persuasion frontier profile is non-improving at F^* , which is (A.3). The final statement follows from Proposition 3. $\square$

B Proofs for applications

The proofs below specialize the frontier restriction and the moment-problem representation to the political-narrative application in Section 4.

B.1 Proof of Proposition 4

Proof of Proposition 4. Part (i). Fix $t \in (0, 1)$ and let X denote the random type. The pointwise inequality

$$X \leq t + (1 - t)\mathbf{1}\{X > t\}$$

implies

$$F((t, 1]) \geq \frac{\hat{\theta} - t}{1 - t},$$

and this lower bound is informative only when $\hat{\theta} > t$. A second pointwise inequality,

$$X(X - t) \leq (1 - t)\mathbf{1}\{X > t\},$$

implies

$$F((t, 1]) \geq \frac{\mathbb{E}_F[X^2] - t\hat{\theta}}{1 - t} \geq \frac{\underline{\sigma}^2 - \hat{\theta}(t - \hat{\theta})}{1 - t}.$$

Together with nonnegativity of tail mass, these two inequalities give the lower bound in (14).

The bound is tight. If $\hat{\theta} \leq t$ and $\underline{\sigma}^2 \leq \hat{\theta}(t - \hat{\theta})$, the distribution $(1 - \hat{\theta}/t)\delta_0 + (\hat{\theta}/t)\delta_t$ is feasible and has tail mass 0. If $\hat{\theta} > t$ and $\underline{\sigma}^2 \leq (\hat{\theta} - t)(1 - \hat{\theta})$, the distribution $((1 - \hat{\theta})/(1 - t))\delta_t + ((\hat{\theta} - t)/(1 - t))\delta_1$ is feasible and has tail mass $(\hat{\theta} - t)/(1 - t)$. In the remaining case, use $a\delta_0 + b\delta_t + c\delta_1$, where

$$b = \frac{\hat{\theta}(1 - \hat{\theta}) - \underline{\sigma}^2}{t(1 - t)}, \quad c = \frac{\underline{\sigma}^2 - \hat{\theta}(t - \hat{\theta})}{1 - t}, \quad a = 1 - b - c.$$

A direct calculation gives mean $\hat{\theta}$, variance $\underline{\sigma}^2$, and tail mass c . Nonnegativity follows from $\underline{\sigma}^2 < \hat{\theta}(1 - \hat{\theta})$, the remaining-case inequality, and $a = (\underline{\sigma}^2 + (t - \hat{\theta})(1 - \hat{\theta}))/t \geq 0$. Thus each piece of (14) is attained.

Part (ii). At $\tau = 1/2$, the boundary menu $(0, 1/2)$ pays w_h to every type and w_l to types above $1/2$, giving payoff guarantee $w_h + w_l \underline{T}(1/2)$ by part (i); the value at $(1/2, 0)$ is symmetric. Their difference $(w_h - w_l)(1 - \underline{T}(1/2))$ has $\underline{T}(1/2) < 1$ since each piece of (14) at $t = 1/2$ is strictly below one under $\underline{\sigma}^2 < \hat{\theta}(1 - \hat{\theta})$. Hence $(0, 1/2)$ weakly dominates iff $w_h \geq w_l$; when $w_h < w_l$ the inequality reverses and the symmetric boundary value is $w_l + w_h \underline{T}(1/2)$ at $(1/2, 0)$. $\square$

B.2 Proof of Proposition 5

Proof of Proposition 5. We prove the case $w_h \geq w_l$; the case $w_h < w_l$ follows by interchanging h and l . By Corollary 3, it is enough to compare the boundary menu $(0, 1/2)$ with h -first frontier menus. Every frontier interior menu has a strictly positive lower threshold, so the Bernoulli distribution $(1 - \hat{\theta})\delta_0 + \hat{\theta}\delta_1$ is feasible and gives payoff $(w_h + w_l)\hat{\theta}$. Hence every frontier interior menu has payoff guarantee at most $(w_h + w_l)\hat{\theta}$.

The boundary value is $w_h + w_l \underline{T}(1/2)$. If $\hat{\theta} \leq 1/2$, then $w_h + w_l \underline{T}(1/2) \geq w_h \geq (w_h + w_l)/2 \geq (w_h + w_l)\hat{\theta}$. If $\hat{\theta} > 1/2$, then Proposition 4 gives $\underline{T}(1/2) \geq 2\hat{\theta} - 1$, and therefore

$$w_h + w_l \underline{T}(1/2) - (w_h + w_l)\hat{\theta} \geq (w_h - w_l)(1 - \hat{\theta}) \geq 0.$$

Thus $(0, 1/2)$ weakly dominates every frontier interior menu. Together with Proposition 4, it is an optimal menu when $w_h \geq w_l$; the symmetric boundary menu is optimal when $w_h < w_l$. $\square$

B.3 Proof of Proposition 6

Proof of Proposition 6. Part (i) follows from set inclusion: $\underline{\sigma}_1^2 \leq \underline{\sigma}_2^2$ implies $\Pi(\hat{\theta}, \underline{\sigma}_2^2) \subseteq \Pi(\hat{\theta}, \underline{\sigma}_1^2)$, and taking the infimum of a fixed integrand over the smaller set cannot decrease its value; the supremum over menus preserves the inequality. For Part (ii), on the variance-binding term Proposition 4 gives $\underline{T}(1/2) = 2(\underline{\sigma}^2 - \hat{\theta}(1/2 - \hat{\theta}))$ with $\underline{\sigma}^2$ -derivative 2, so by Proposition 4 and Proposition 5 the optimal value has derivative $2 \min\{w_h, w_l\} > 0$. $\square$

B.4 Proof of Proposition 7

Proof of Proposition 7. At $t = 1/2$, Proposition 4 gives

$$\underline{T}(1/2) = \max\{2 \max\{\hat{\theta} - 1/2, 0\}, 2(\underline{\sigma}^2 - \hat{\theta}(1/2 - \hat{\theta})), 0\}.$$

By Proposition 5, the optimal value equals $\max\{w_h, w_l\} + \min\{w_h, w_l\}\underline{T}(1/2)$ throughout. Since $\min\{w_h, w_l\} > 0$, the optimal value has the same monotonicity and minimizers as $\underline{T}(1/2)$.

The variance-binding term has derivative $4\hat{\theta} - 1$. The feasible mean interval has lower endpoint $(1 - \sqrt{1 - 4\underline{\sigma}^2})/2$, which is below $1/4$ when $\underline{\sigma}^2 < 3/16$; hence the decreasing portion exists in that case. At $\hat{\theta} = 1/4$, the variance-binding term equals $2\underline{\sigma}^2 - 1/8$. Therefore the minimum is $\max\{w_h, w_l\}$ if $\underline{\sigma}^2 \leq 1/16$, is $\max\{w_h, w_l\} + \min\{w_h, w_l\}(2\underline{\sigma}^2 - 1/8)$ if $1/16 < \underline{\sigma}^2 < 3/16$, and the value is strictly increasing in $\hat{\theta}$ when $\underline{\sigma}^2 \geq 3/16$. $\square$

B.5 Derivation of the saddle-point condition

Suppose $w_h \geq w_l$, so $M_{h\text{-bd}} = (0, 1/2)$ is robust optimal. By Proposition 4(i), the payoff guarantee $w_h + w_l \underline{T}(1/2)$ is attained by a distribution F^* supported on $\{0, 1/2, 1\}$ with probability masses

$$p_0 = 1 - 2\hat{\theta} + \underline{T}(1/2), \quad p_{1/2} = 2(\hat{\theta} - \underline{T}(1/2)), \quad p_1 = \underline{T}(1/2).$$

Specializing Proposition A.5 to $S = \{h, l\}$, the boundary menu is benchmark-optimal at F^* iff every h -first frontier profile $(x, \theta_l(x))$, $x \in (0, 1/4]$, is non-improving. Along the h -first frontier at $\tau = 1/2$,

$$\tilde{v}(\theta; x, \theta_l(x)) - \tilde{v}(\theta; 0, 1/2) = w_l \mathbf{1}\{\theta_l(x) < \theta \leq 1/2\} - w_h \mathbf{1}\{0 \leq \theta \leq x\}.$$

The least favorable distribution has support $\{0, 1/2, 1\}$. Hence every $x \in (0, 1/4]$ is non-improving iff

$$w_l \cdot 2(\hat{\theta} - \underline{T}(1/2)) \leq w_h(1 - 2\hat{\theta} + \underline{T}(1/2)). \quad (\text{B.1})$$

The reverse strict inequality gives a strictly profitable frontier deviation. The case $w_l > w_h$ is symmetric.

B.6 Boundary optimality under general action thresholds

The political-narrative application of the main paper takes $\tau = 1/2$, where a boundary menu is globally optimal (Proposition 5) and the optimal-value comparative statics of Propositions 6 and 7 apply throughout. At arbitrary $\tau \in (0, 1)$, the tail-mass identity (14) of Proposition 4 continues to determine the payoff of any boundary menu. The following proposition gives primitive sufficient conditions for boundary optimality and records the comparative statics on those regions. In particular, the low-mean negative effect of raising the average prior survives for every τ because it lies inside an automatically boundary-optimal region.

Proposition B.1 (Boundary optimality under general action thresholds)

Fix $\tau \in (0, 1)$, $0 < \hat{\theta} < 1$, and $0 < \underline{\sigma}^2 < \hat{\theta}(1 - \hat{\theta})$, with the mean-variance ambiguity set $\Pi(\hat{\theta}, \underline{\sigma}^2)$ of Proposition 4. Let $\underline{T}(\tau)$ denote the value of (14) at $t = \tau$.

(i) The boundary menus have payoff guarantees

$$V_h^{\text{bd}} = w_h + w_l \underline{T}(\tau), \quad V_l^{\text{bd}} = w_l + w_h \underline{T}(\tau).$$

(ii) Suppose $w_h \geq w_l$. The boundary menu $M_{h\text{-bd}}$ is optimal whenever

$$\frac{w_h}{w_l} \geq \frac{\min\{\hat{\theta}, \tau\}}{1 - \min\{\hat{\theta}, \tau\}}. \quad (\text{B.2})$$

If (B.2) fails and the variance piece of $\underline{T}(\tau)$ is active, $M_{h\text{-bd}}$ is still optimal whenever

$$\underline{\sigma}^2 \geq (1 - \hat{\theta}) \left[\hat{\theta} - (1 - \tau) \frac{w_h}{w_l} \right]. \quad (\text{B.3})$$

In particular, when $\hat{\theta} \leq 1/2$ a boundary menu is optimal for every feasible $\underline{\sigma}^2$: $M_{h\text{-bd}}$ if $w_h \geq w_l$, $M_{l\text{-bd}}$ if $w_l > w_h$.

(iii) The optimal value is weakly increasing in $\underline{\sigma}^2$. On any boundary-optimal region where the variance piece of $\underline{T}(\tau)$ is active, it is strictly increasing with derivative $\min\{w_h, w_l\}/(1 - \tau)$.

(iv) Let $C := \max\{w_h, w_l\}$ and $m := \min\{w_h, w_l\}$. At every feasible point satisfying

$$\hat{\theta} < \frac{\tau}{2} \quad \text{and} \quad \underline{\sigma}^2 > \hat{\theta}(\tau - \hat{\theta}), \quad (\text{B.4})$$

the relevant boundary menu is optimal, the variance piece of $\underline{T}(\tau)$ is active, and the optimal value satisfies

$$V^* = C + \frac{m(\underline{\sigma}^2 - \hat{\theta}(\tau - \hat{\theta}))}{1 - \tau}, \quad \frac{\partial V^*}{\partial \hat{\theta}} = \frac{m(2\hat{\theta} - \tau)}{1 - \tau} < 0.$$

Thus raising the average prior lowers the actual robust value on this low-mean variance-binding region.

For fixed $\underline{\sigma}^2$, this decreasing region is nonempty if and only if

$$\underline{\sigma}^2 < \frac{\tau(2 - \tau)}{4}. \quad (\text{B.5})$$

More precisely, with $\hat{\theta}_{\text{lo}} := \frac{1 - \sqrt{1 - 4\underline{\sigma}^2}}{2}$, the optimal value is strictly decreasing on

$$\left(\hat{\theta}_{\text{lo}}, \bar{\theta}_\tau(\underline{\sigma}^2) \right), \quad \bar{\theta}_\tau(\underline{\sigma}^2) := \begin{cases} \frac{\tau - \sqrt{\tau^2 - 4\underline{\sigma}^2}}{2}, & \underline{\sigma}^2 < \tau^2/4, \\ \tau/2, & \underline{\sigma}^2 \geq \tau^2/4. \end{cases}$$

At $\tau = 1/2$, Proposition 5 makes the boundary-optimal region cover the entire feasible mean range, so (B.5) becomes $\underline{\sigma}^2 < 3/16$ with turning point $\hat{\theta} = 1/4$, and Proposition 7 is recovered.

Proof. *Part (i).* The boundary menu $(0, \tau)$ yields per-type payoff $\bar{u}_0 + w_h + w_l \mathbf{1}\{\theta > \tau\}$ under the lower semi-continuous envelope, so its payoff guarantee over $\Pi(\hat{\theta}, \underline{\sigma}^2)$ is $w_h + w_l \underline{T}(\tau)$ by Proposition 4. The formula for V_l^{bd} follows by interchanging h and l . Since $\underline{T}(\tau) \leq 1$, $V_h^{\text{bd}} - V_l^{\text{bd}} = (w_h - w_l)(1 - \underline{T}(\tau)) \geq 0$ when $w_h \geq w_l$.

Part (ii). We prove the case $w_h \geq w_l$; the case $w_l > w_h$ is symmetric. By Corollary 3, it suffices to compare $M_{h\text{-bd}}$ with h -first frontier interior menus. Every interior menu has positive lower threshold, so the Bernoulli distribution $(1 - \hat{\theta})\delta_0 + \hat{\theta}\delta_1$ lies in $\Pi(\hat{\theta}, \underline{\sigma}^2)$ (variance $\hat{\theta}(1 - \hat{\theta}) > \underline{\sigma}^2$) and gives expected payoff $(w_h + w_l)\hat{\theta}$, bounding the payoff guarantee of every interior menu from above by $(w_h + w_l)\hat{\theta}$. Thus $M_{h\text{-bd}}$ is optimal whenever

$$w_h + w_l \underline{T}(\tau) \geq (w_h + w_l)\hat{\theta}. \quad (\text{B.6})$$

If $\hat{\theta} \leq \tau$, (B.6) simplifies via $\underline{T}(\tau) \geq 0$ to $w_h \geq (w_h + w_l)\hat{\theta}$, equivalent to $w_h/w_l \geq \hat{\theta}/(1 - \hat{\theta})$. If $\hat{\theta} > \tau$, (14) gives $\underline{T}(\tau) \geq (\hat{\theta} - \tau)/(1 - \tau)$, and substituting yields $w_h/w_l \geq \tau/(1 - \tau)$ for (B.6). The two cases combine into (B.2).

When (B.2) fails and the variance piece of $\underline{T}(\tau)$ is active, $\underline{T}(\tau) = (\underline{\sigma}^2 - \hat{\theta}(\tau - \hat{\theta}))/ (1 - \tau)$, and substituting into (B.6) and using $\hat{\theta}(\tau - \hat{\theta}) + (1 - \tau)\hat{\theta} = \hat{\theta}(1 - \hat{\theta})$ yields (B.3). For $\hat{\theta} \leq 1/2$, $\min\{\hat{\theta}, \tau\} \leq 1/2$ gives $\min\{\hat{\theta}, \tau\}/(1 - \min\{\hat{\theta}, \tau\}) \leq 1 \leq w_h/w_l$, so (B.2) holds; the case $w_l > w_h$ gives $M_{l\text{-bd}}$ symmetrically.

Part (iii). Weak monotonicity in $\underline{\sigma}^2$ follows from set inclusion: $\underline{\sigma}_1^2 \leq \underline{\sigma}_2^2$ gives $\Pi(\hat{\theta}, \underline{\sigma}_2^2) \subseteq \Pi(\hat{\theta}, \underline{\sigma}_1^2)$, so every menu's payoff guarantee is weakly larger at $\underline{\sigma}_2^2$, and the supremum over menus preserves the inequality. On the variance-active boundary-optimal region, differentiating part (i) gives $\partial_{\underline{\sigma}^2} V_h^{\text{bd}} = w_l/(1 - \tau)$ and $\partial_{\underline{\sigma}^2} V_l^{\text{bd}} = w_h/(1 - \tau)$; whichever boundary is optimal, the active coefficient is $\min\{w_h, w_l\}$.

Part (iv). If (B.4) holds, then $\hat{\theta} < \tau/2 < 1/2$, so part (ii) implies boundary optimality for the boundary menu associated with the larger payoff weight. Since $\hat{\theta} < \tau$, the mean-forced term in (14) is zero, and the strict inequality $\underline{\sigma}^2 > \hat{\theta}(\tau - \hat{\theta})$ makes the variance term active:

$$\underline{T}(\tau) = \frac{\underline{\sigma}^2 - \hat{\theta}(\tau - \hat{\theta})}{1 - \tau}.$$

Substituting this expression into the boundary value in part (i) gives the value with coefficient $m = \min\{w_h, w_l\}$, and differentiating gives $m(2\hat{\theta} - \tau)/(1 - \tau) < 0$.

For fixed $\underline{\sigma}^2$, feasibility requires $\hat{\theta} \in (\hat{\theta}_{\text{lo}}, \hat{\theta}_{\text{hi}})$, where

$$\hat{\theta}_{\text{lo}} = \frac{1 - \sqrt{1 - 4\underline{\sigma}^2}}{2}, \quad \hat{\theta}_{\text{hi}} = \frac{1 + \sqrt{1 - 4\underline{\sigma}^2}}{2}.$$

A feasible point satisfying $\hat{\theta} < \tau/2$ exists if and only if $\hat{\theta}_{\text{lo}} < \tau/2$, which is equivalent to (B.5). On $[0, \tau/2]$, the function $\hat{\theta}(\tau - \hat{\theta})$ is increasing. Hence the additional variance-activity requirement $\underline{\sigma}^2 > \hat{\theta}(\tau - \hat{\theta})$ holds below $(\tau - \sqrt{\tau^2 - 4\underline{\sigma}^2})/2$ when $\underline{\sigma}^2 < \tau^2/4$, and holds throughout $\hat{\theta} < \tau/2$ when $\underline{\sigma}^2 \geq \tau^2/4$. This yields the interval in the statement. The specialization $\tau = 1/2$ gives $\tau(2 - \tau)/4 = 3/16$ and turning point $\hat{\theta} = 1/4$. $\square$

Remark on general- τ menu classification. Outside the sufficient boundary-optimality regions above, frontier-interior menus may dominate. When $w_h \geq w_l$, the h -first frontier can be

parameterized by $x = \theta_h \in [0, \tau/2]$ with

$$\theta_l(x) = \frac{\tau - (1 + \tau)x}{1 - 2x}.$$

For each fixed $x > 0$, Nature's problem is a finite moment problem over the four endpoints $\{0, x, \theta_l(x), 1\}$. The active support, and hence the formula for the inner value, changes with $(\hat{\theta}, \underline{\sigma}^2, x, w_h/w_l)$. Classifying all primitive cases would require comparing several frontier-interior cases against the boundary value. For this reason, the proposition records sufficient conditions and the low-mean comparative static.

C Product reviews and an ambiguous consumer prior

This application reads the model through the single-receiver interpretation introduced in Section 2. A seller wants a consumer to buy a product ($a = 1$), while the consumer buys only when quality is high ($\omega = H$). The consumer's prior θ that quality is high is privately known, and the seller knows only that it is drawn from a distribution F whose mean lies in an interval $[\underline{\theta}, \bar{\theta}]$. Public reviews generate h or l , with $0 < \pi_L < \pi_H < 1$. I normalize the seller's utility from no purchase to 0, so $\bar{u}_0 = 0$, and let w_h and w_l denote her expected revenue from a purchase induced after a positive and a negative review.

Let $0 < \underline{\theta} \leq \bar{\theta} < 1$ and define

$$\Pi([\underline{\theta}, \bar{\theta}]) := \{F \in \Delta([0, 1]) : \underline{\theta} \leq \mathbb{E}_F[\theta] \leq \bar{\theta}\}.$$

The question is which menu achieves the highest payoff guarantee: a boundary menu targeting one realization, or an interior menu that spreads persuasion across both realizations.

Proposition C.1 (Menu selection under interval mean ambiguity)

Assume $0 < \underline{\theta} \leq \bar{\theta} < 1$ and $\tau \in (0, 1)$.

(i) For every menu M ,

$$\min_{F \in \Pi([\underline{\theta}, \bar{\theta}])} \int \tilde{v}(\theta; M) dF = \min_{\substack{F \in \Delta([0, 1]) \\ \mathbb{E}_F[\theta] = \underline{\theta}}} \int \tilde{v}(\theta; M) dF.$$

Thus the interval-mean robust problem coincides with the point-mean problem at the lower endpoint $\underline{\theta}$. This is the specialization of Proposition 2 to $g(\theta) = \theta$, $Y = [\underline{\theta}, \bar{\theta}]$: the non-decreasing per-type payoff $\tilde{v}(\cdot; M)$ forces the worst-case mean to the lower endpoint.

(ii) Suppose $w_h \geq w_l$. The boundary menu $M_{h\text{-bd}}$ with threshold pair $(0, \tau)$ is optimal iff

$$\frac{w_h}{w_h + w_l} \geq \begin{cases} 2\underline{\theta} - \tau, & \underline{\theta} \leq \tau, \\ \tau, & \underline{\theta} > \tau. \end{cases}$$

When it is optimal, its value is $w_h + w_l \max\{(\underline{\theta} - \tau)/(1 - \tau), 0\}$. When this condition fails, an optimal h -first frontier interior menu has threshold pair

$$\theta_h = \frac{\tau - \frac{w_h}{w_h + w_l}}{2 \left(1 - \frac{w_h}{w_h + w_l}\right)}, \quad \theta_l = \frac{\tau + \frac{w_h}{w_h + w_l}}{2},$$

and its value is $(w_h + w_l)(\underline{\theta} - \theta_h)/(1 - \theta_h)$.

- (iii) If $w_l \geq w_h$, the symmetric statement holds after interchanging h and l together with w_h and w_l : $M_{l\text{-bd}}$ replaces $M_{h\text{-bd}}$, the cutoff uses $w_l/(w_h + w_l)$, when it is optimal its value is $w_l + w_h \max\{(\underline{\theta} - \tau)/(1 - \tau), 0\}$, and the optimal interior menu is the corresponding l -first frontier menu with value obtained from (ii) by exchanging w_h and w_l .

Part (i) follows because $\tilde{v}(\cdot; M)$ is non-decreasing in type for every menu M . The seller is rewarded only by what the data rules out from below: tightening $\bar{\theta}$ has no value here, while raising $\underline{\theta}$ can raise the guarantee. Parts (ii) and (iii) compare concentrating persuasion at one endpoint with spreading it along the frontier. A boundary menu is optimal when the insured realization is valuable enough, or when the lower-endpoint mean is low enough that Nature can keep little mass above the other threshold. When the condition fails, the seller uses both review realizations: lowering the less protected threshold is worth raising the fully insured one, and the optimal interior pair is where the one-moment lower envelope switches from the middle step to the chord that reaches the top step (see Figure 2).

Proof of Proposition C.1. For every menu M , the lower semi-continuous payoff $\tilde{v}(\theta; M)$ is non-decreasing in θ . Take any $F \in \Pi([\underline{\theta}, \bar{\theta}])$ with $\mathbb{E}_F[\theta] > \underline{\theta}$ and replace it by $(\underline{\theta}/\mathbb{E}_F[\theta])F + (1 - \underline{\theta}/\mathbb{E}_F[\theta])\delta_0$. The new distribution has mean $\underline{\theta}$ and remains in $\Pi([\underline{\theta}, \bar{\theta}])$. Since $\tilde{v}(0; M) \leq \tilde{v}(\theta; M)$ for every θ , this replacement weakly lowers the expected payoff. Hence it is without loss to restrict Nature's minimization to distributions with mean $\underline{\theta}$. The reverse inequality is immediate because the point-mean set is contained in $\Pi([\underline{\theta}, \bar{\theta}])$.

This proves part (i). For parts (ii) and (iii), we prove the case $w_h \geq w_l$; the case $w_l \geq w_h$ follows by interchanging h and l together with w_h and w_l .

By Corollary 3, it is enough to optimize over the h -first frontier. Write

$$W := w_h + w_l, \quad p := \frac{w_h}{W} \in [1/2, 1).$$

Parameterize the h -first frontier by $x = \theta_h \in [0, \tau/2]$ and

$$y(x) := \theta_l(x) = \frac{\tau - (1 + \tau)x}{1 - 2x}, \quad d(x) := \frac{y(x) - x}{1 - x} = \frac{\tau - 2x}{1 - 2x}.$$

Then $d(x)$ decreases from τ to 0. For $x > 0$, the lower semi-continuous payoff is 0 on $[0, x]$, w_h on $(x, y(x)]$, and W on $(y(x), 1]$. The one-moment convexification gives the point-mean frontier value $\Phi_x(\underline{\theta})$:

$$\Phi_x(\underline{\theta}) = \begin{cases} 0, & \underline{\theta} \leq x, \\ Wp \frac{\underline{\theta} - x}{y(x) - x}, & x < \underline{\theta} \leq y(x), \\ W \left[p + (1 - p) \frac{\underline{\theta} - y(x)}{1 - y(x)} \right], & \underline{\theta} > y(x), \end{cases} \quad \text{if } d(x) \geq p,$$

and

$$\Phi_x(\underline{\theta}) = \begin{cases} 0, & \underline{\theta} \leq x, \\ W \frac{\underline{\theta} - x}{1 - x}, & \underline{\theta} > x, \end{cases} \quad \text{if } d(x) \leq p.$$

At equality the two formulas agree.

First suppose $p \geq \tau$. Then $d(x) \leq p$ for every x . The expression $(\underline{\theta} - x)/(1 - x)$ is decreasing in x whenever $x < \underline{\theta}$, so every interior frontier value is at most $W\underline{\theta}$. The boundary value is

$$B := W \left[p + (1 - p) \frac{(\underline{\theta} - \tau)_+}{1 - \tau} \right].$$

If $\underline{\theta} \leq \tau$, then $B = Wp \geq W\tau \geq W\underline{\theta}$. If $\underline{\theta} > \tau$, direct algebra gives $B \geq W\underline{\theta}$. Thus the boundary menu is optimal for all $\underline{\theta}$ when $p \geq \tau$.

Now suppose $p < \tau$. Let x_p be the unique solution to $d(x) = p$:

$$x_p = \frac{\tau - p}{2(1 - p)}, \quad y_p := y(x_p) = \frac{\tau + p}{2}.$$

For algebraic comparisons, write

$$K := W \frac{\underline{\theta} - x_p}{1 - x_p}$$

This is the value at the frontier point x_p when $\underline{\theta} > x_p$; when $\underline{\theta} \leq x_p$, that point yields value zero. For $\underline{\theta} \leq \tau$, the boundary condition $p \geq 2\underline{\theta} - \tau$ is equivalent to $\underline{\theta} \leq y_p$. In that case $B = Wp$, and the inequality $\underline{\theta} \leq y_p$ gives $K \leq B$. If $x \geq x_p$ and $\underline{\theta} \leq x$, then $\Phi_x(\underline{\theta}) = 0 \leq B$. If $x \geq x_p$ and $x < \underline{\theta}$, the chord formula applies and is decreasing in x , so $\Phi_x(\underline{\theta}) \leq K \leq B$. If $x < x_p$, then $y(x) > y_p \geq \underline{\theta}$. The remaining case is $x < \underline{\theta} \leq y(x)$, and direct algebra gives

$$\frac{B - \Phi_x(\underline{\theta})}{W} = \frac{p[\tau - \underline{\theta} - x(1 + \tau - 2\underline{\theta})]}{(\tau - 2x)(1 - x)}.$$

The denominator is positive. Since the bracket is decreasing in x and nonnegative at $x = x_p$ under the boundary condition, this expression is nonnegative. Hence the boundary menu weakly dominates the entire frontier whenever the stated boundary condition holds on $\underline{\theta} \leq \tau$.

If $p < \tau$ and the boundary condition fails, then $\underline{\theta} > y_p$: for $\underline{\theta} \leq \tau$ this is equivalent to $p < 2\underline{\theta} - \tau$, and for $\underline{\theta} > \tau$ it holds automatically. The frontier value is then maximized at x_p . Indeed, for $x \geq x_p$, the chord formula is decreasing in x , so $\Phi_x(\underline{\theta}) \leq K$. For $x < x_p$ and $\underline{\theta} > y(x)$, the third line of the convex-envelope formula is increasing in x , so again $\Phi_x(\underline{\theta}) \leq K$. For $x < x_p$ and $\underline{\theta} \leq y(x)$,

$$\frac{K - \Phi_x(\underline{\theta})}{W} = \frac{[\tau - p - 2(1 - p)x][\underline{\theta}(2 - p - 2x) + x(p + \tau) - \tau]}{(\tau - 2x)(1 - x)(2 - p - \tau)}.$$

The denominator is positive, and the first bracket is positive because $x < x_p$. The second bracket is increasing in $\underline{\theta}$ and, at $\underline{\theta} = y_p$, equals $p(2 - p - \tau)/2 > 0$. Hence $K > \Phi_x(\underline{\theta})$ in this case as well.

Combining the cases gives the boundary condition in part (ii). When it fails, the optimal h -first frontier interior menu is (x_p, y_p) , which gives the displayed threshold pair and value. $\square$

Corollary C.1 (Comparative statics under interval mean ambiguity)

Assume $0 < \underline{\theta} \leq \bar{\theta} < 1$ and $\tau \in (0, 1)$.

- (i) *The optimal value and the optimal-menu set do not depend on $\bar{\theta}$.*
- (ii) *The optimal value is weakly increasing in $\underline{\theta}$.*
- (iii) *Suppose $w_h \geq w_l$. If $w_h/(w_h + w_l) \geq \tau$, then $M_{h\text{-bd}}$ is optimal for every $\underline{\theta}$. If $w_h/(w_h + w_l) < \tau$, then $M_{h\text{-bd}}$ is optimal when $\underline{\theta} \leq (w_h/(w_h + w_l) + \tau)/2$, and an interior h -first frontier menu is optimal above that cutoff. The case $w_l \geq w_h$ is symmetric.*
- (iv) *In the h -first interior region, increasing $w_h/(w_h + w_l)$ lowers θ_h and raises θ_l , while increasing τ raises both thresholds. The l -first interior region is symmetric.*

Corollary C.1 describes how the lower mean bound and payoff weights move the robust menu. Raising $\underline{\theta}$ weakly improves the payoff guarantee because Nature's least favorable prior distribution must be more optimistic about quality. If the larger payoff weight is at least τ , the seller insures the high-payoff review realization for every $\underline{\theta}$; otherwise this boundary menu remains optimal only at low $\underline{\theta}$, and the seller moves into the frontier interior once the data rules out sufficiently pessimistic prior distributions. In the interior region, increasing the payoff weight of the more valuable realization lowers that realization's threshold and raises the other, while a larger τ raises both thresholds.

Proof of Corollary C.1. Part (i) follows from Proposition C.1(i): for each menu M , the payoff guarantee equals the point-mean problem at $\underline{\theta}$, which contains no $\bar{\theta}$.

For part (ii), take $0 < \underline{\theta}_1 < \underline{\theta}_2 < 1$. For any menu M and any $F \in \Delta([0, 1])$ with $\mathbb{E}_F[\theta] = \underline{\theta}_2$, the distribution $(\underline{\theta}_1/\underline{\theta}_2)F + (1 - \underline{\theta}_1/\underline{\theta}_2)\delta_0$ has mean $\underline{\theta}_1$. Since $\tilde{v}(\theta; M)$ is non-decreasing in θ , the expected payoff under this new distribution is weakly below the expected payoff under F . Hence the point-mean payoff guarantee for each fixed menu is weakly increasing in $\underline{\theta}$. Maximizing over menus preserves the inequality.

For part (iii), suppose $w_h \geq w_l$. The condition in Proposition C.1(ii) is equivalent to $\underline{\theta} \leq (w_h/(w_h + w_l) + \tau)/2$ on $\underline{\theta} \leq \tau$ and to $w_h/(w_h + w_l) \geq \tau$ on $\underline{\theta} > \tau$. When $w_h/(w_h + w_l) \geq \tau$, the boundary optimality condition is satisfied for every $\underline{\theta}$, so $M_{h\text{-bd}}$ is always optimal. When $w_h/(w_h + w_l) < \tau$, $(w_h/(w_h + w_l) + \tau)/2 < \tau$ pins down the boundary-optimality range; beyond it, Proposition C.1(ii) gives the h -first frontier interior menu.

For part (iv), in the h -first interior region the thresholds are those in Proposition C.1(ii). Differentiating with respect to $w_h/(w_h + w_l)$ gives $(\tau - 1)/(2(1 - w_h/(w_h + w_l))^2) < 0$ for θ_h and $1/2 > 0$ for θ_l . Differentiating with respect to τ gives $1/(2(1 - w_h/(w_h + w_l))) > 0$ for θ_h and $1/2 > 0$ for θ_l . $\square$

This differs from Carrasco et al. (2018), where a first-moment restriction alone does not guarantee strictly positive worst-case revenue. Here, because the receiver's type lies in the compact set $[0, 1]$ and payoffs are bounded step functions of persuasion thresholds, the same kind of first-moment information limits how much mass Nature can move below the relevant thresholds and can therefore secure a strictly positive payoff above $\bar{u}_0$.

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